Understanding the tomato-cutting project

Expanded edition: history and mechanics explained

From the original research ambition to cutting-model development, robot integration and vision-guided knife pickup.

Record reviewed through 14 November 2026.
Prepared for the project researcher.
Robotics basics assumed; simulation and learning explained step by step.

Read this as an explanation of the evidence, not a claim that the entire system is finished. The latest pickup work was still in progress at the review cutoff.

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How to read this expanded edition

Chapters 1–18 preserve the chronological project story. Chapters 19–44 slow down to explain the simulation, tomato representations, numerical failures and interpretation of results. The glossary and evidence map follow. Diagrams are explanatory schematics; worked arithmetic is labeled as illustrative rather than measured project data. This revision expands the explanation of the reviewed record, without claiming new experiments after the 14 November cutoff.

Understanding the tomato-cutting project

A step-by-step history of the work, the failures, and the decisions

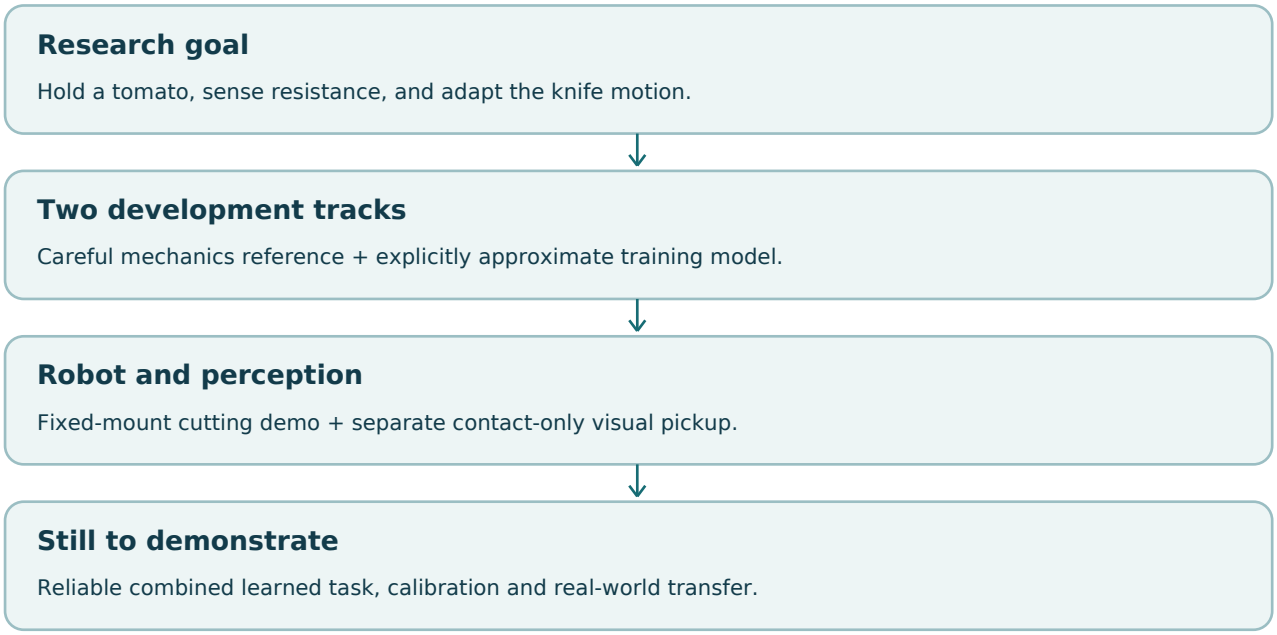
This document is for you: a readable explanation of how the project developed, rather than a paper claiming that everything is finished. It covers the accessible research record from the early October 2026 planning documents through the November 14 repository update and latest pickup-chat messages inspected for this report.

The story has several branches. Robot setup, knife grasping, cutting physics, numerical accuracy, performance, visual appearance and learning developed at different speeds. A successful result in one branch did not automatically complete the others. Keeping those branches separate is the key to understanding the project.

The account is reconstructed from saved plans, result reports, experiment folders, the repository and the continuation chat. Some early steps have only summary records. Dates on older living documents sometimes reflect when the document began, not every later edit. Where the sequence is known but the exact date is not, this document says so. It does not pretend that every command or abandoned thought can be recovered.

Source labels in square brackets lead to the evidence appendix. That appendix also contains an index of individual experiment reports and plain-language explanations of the later version sequence. The underlying reports remain the authority for exact settings and acceptance limits.

1. The project in one view



1.1 The original ambition

The original ambition was a two-arm robot that could hold a tomato and slice it intelligently. One arm would stabilize the food. The other would move the knife. The controller would use camera information and contact or force information to adjust to uncertainty: a tomato may be soft, firm, slippery, differently shaped, or positioned unexpectedly. [S01]

The early research question was deliberately ambitious. It combined hierarchical residual reinforcement learning, multiple sensor types, compliant control and transfer from simulation to a real robot. These were proposed research objectives. They were not capabilities already demonstrated by the robot.

1.2 What we have now

The project now has a large body of source code, diagnostics, simulation scenes, experiment reports and preserved failure evidence. It has an approximate tomato model that completed restricted simulated cutting tasks, a fixed-mount robot/kitchen cutting demonstration, and a separate camera-guided contact-only knife pickup prototype. Those are important but different achievements. [S12-S16]

The stricter cutting-physics branch remains an unfinished numerical research effort. The approximate model deliberately accepts material and discretization limitations. The pickup branch has a strong first result on a limited knife setup, but broader shapes and

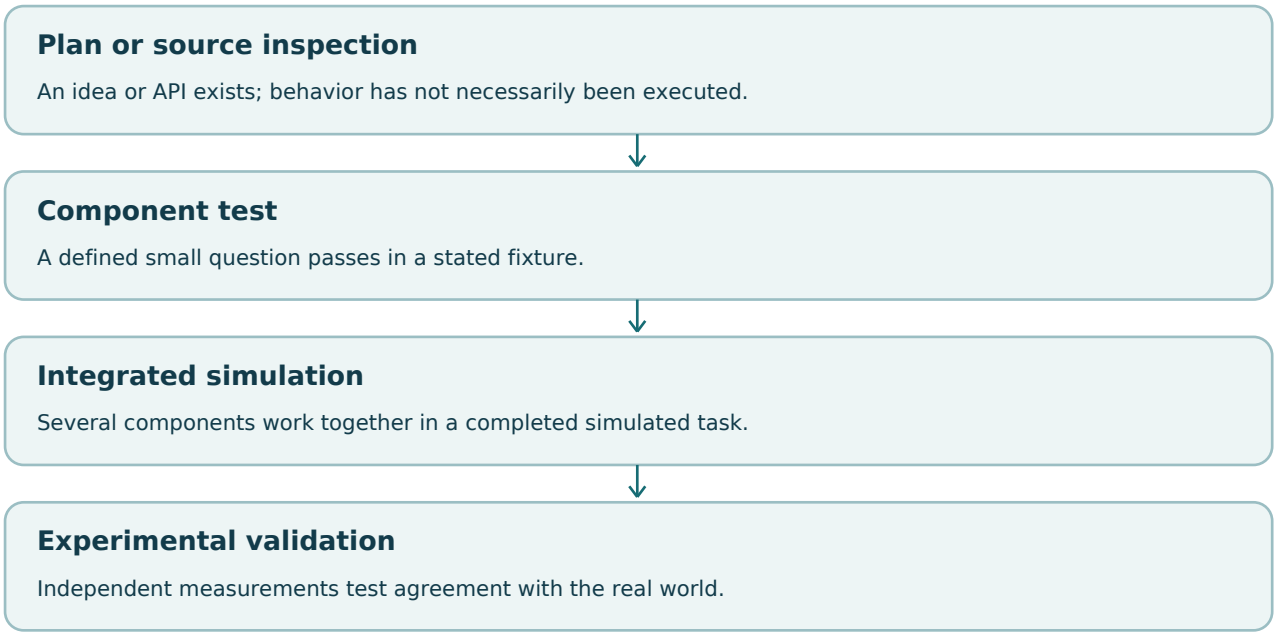
resting poses remain under development. A complete learned, calibrated, dual-arm real-world tomato-cutting system is not established.

1.3 Why the work looked like repeated detours

The project repeatedly exposed a hidden prerequisite. Before learning cutting, the material had to separate correctly. Before trusting separation, contact and energy had to be checked. Before running many learning episodes, the simulator had to be fast enough. Before asking a hand to carry the knife, the grip had to work physically. These were dependencies, not interchangeable checklist items.

However, the history also shows a real scope problem: trying to make a very accurate custom cutting simulator and a sophisticated robot-learning system at the same time created many possible research projects. Later decisions increasingly separated an accurate reference track from an approximate training track.

2. How to read the evidence



2.1 A plan is not a result

A plan describes intended work. Implemented code means an attempt exists. A unit test checks a small defined behavior. A completed simulation demonstrates behavior in that simulation. A calibrated physical experiment tests agreement with the real world. These are different levels of evidence.

For example, a test can prove that an energy counter updates correctly without proving that the knife actually produces a realistic cut. A scene can show a knife in a hand without

proving that finger friction supports it. The report will keep making these distinctions because they explain many apparent contradictions in earlier updates.

2.2 A coupon is a small test specimen

The project often tested a small block, interface or pair of pieces instead of a whole tomato. This is called a coupon. It isolates one question: whether contact pushes in the right direction, whether a damaged interface stays damaged, or whether one fragment can move without dragging another.

Passing a coupon is useful. It is easier to diagnose than a full robot scene. But the whole system adds shape, support, contact combinations, longer time histories and more computation. A coupon pass is a building block, not a full-system certificate.

2.3 Three meanings of a successful cut

A geometric cut means the representation has been divided along a selected region. A mechanically independent cut means the resulting pieces no longer transmit unintended internal forces. A work-paid physical-model cut also requires a declared fracture rule to spend irreversible work before connections disappear.

Even that last category is still a simulation result. Realistic tomato cutting additionally needs measured material behavior, appropriate blade contact, and experimental validation. The approximate model later achieved work-paid simulated cuts without meeting the stricter continuum-accuracy requirements.

2.4 Why failed runs remain in the record

A failed trial tells us which assumption did not work. Deleting it would make a later improvement look more reliable than it was. The project therefore retained failed configurations, logs, source snapshots and rejected candidate states.

This also protects comparisons. Once a test case has been used to tune a controller, it becomes a development or regression case. It should not later be described as an untouched independent test.

3. Early October: research planning and robot foundations

3.1 The first research architecture

The October 5 decision log placed planning, literature, experiments, simulation, learning, hardware and manuscript work in one Git-based research workspace. Structured research-state files were intended to hold questions, decisions, experiment definitions and claims. [S01]

This was an organizational foundation. It aimed to make the work traceable across long chats and multiple tools. It did not mean that all proposed components had been implemented.

3.2 The proposed arm roles

The early embodiment described ARX AR5-L6 arms and LinkerHand O6 dexterous hands. The left side would hold the tomato and the right side would slice. Later practical work concentrated heavily on the right arm, knife and cutting environment.

The early documents also proposed high-rate tactile or force feedback. Hardware encoder interfaces, sensor variants and calibrated limits were not all verified in the later audit. Therefore, those proposed sensors must not be confused with channels actually available to a trained controller.

3.3 Why residual learning was proposed

Residual learning means starting with an understandable baseline movement and learning a correction. For example, a baseline may move the knife downward while sawing; a learner adjusts speed or compliance when resistance changes.

The attraction was a smaller search problem than asking a network to discover every joint movement from nothing. The early proposed action included Cartesian velocity adjustments and compliance gains. This was a design direction, not evidence that residual cutting had been trained successfully.

3.4 Early platform assumptions needed later verification

The first planning documents selected Isaac Sim, Isaac Lab and PhysX deformables and anticipated efficient GPU training. Some statements in those documents were expectations about cutting and parallelism, not demonstrated installed capabilities.

Later source inspections and numerical experiments showed why those expectations needed testing. A simulator supporting deformable objects does not automatically support the particular fracture law, contact fidelity, persistent cuts and robot coupling required here.

3.5 Joint stability and virtual inertia

An October 7 decision approved added joint armature and default proportional-derivative gains for simulation stability. Armature adds virtual rotational inertia to a joint; it can make numerical motion less explosive when light fingers connect to heavier links. [S01]

This is a modeling choice with a cost. It changes dynamic response, particularly at high frequency. The record explicitly noted a simulation-to-real risk and proposed actuator randomization as a mitigation. Approval of a stable simulated joint is not proof that its real dynamics match.

4. October 19-era audit: could the laptop support the workflow?

4.1 Counting and checking the actual model

The phase audit identified seven independent right-arm joints, six independent hand joints and five passive hand followers. It also checked software and a CUDA smoke test. Tactile channels were excluded where their real interfaces were unverified. [S02]

A follower joint is not another independent command. This distinction matters when defining actions, observations and control limits. Counting visible joints alone can produce the wrong controller interface.

4.2 The resource benchmark

A minimal scene combined the robot, board, countertop, free knife, an RGB-D camera and a small learning workload. RGB-D means a color image plus a depth image. The long run lasted about 30 minutes, completed 277 resets and performed 12 tiny PPO updates. [S02]

PPO is a reinforcement-learning optimization algorithm. In this test, it used a posture reward and an untrained image encoder. The purpose was to exercise software, memory, reset and checkpoint behavior. It did not learn knife recognition, pickup or cutting.

4.3 What the measured numbers meant

The long probe recorded roughly 7.747 GB peak process RAM, 2,658 MiB peak total GPU use and 9.83 actions per wall-clock second. Post-warmup memory growth was small over that measured window. These were encouraging feasibility measurements for one environment, not evidence of multi-day stability or a large parallel training capacity.

A checkpoint is a saved state used to continue work. The benchmark checked restoration of optimizer and random-generator state and a deliberate stop/resume path. It restarted a fresh physical episode; it did not demonstrate exact restoration of every hidden in-flight contact state.

4.4 Safeguards became part of the workflow

Startup reserves, checkpoint reserves, hard memory stops and a shared controller lock were added or verified. The lock prevents two controllers from issuing commands into the same shared setup. These protections were there to preserve the laptop and evidence, not to make a failed physical test pass.

The decision was to continue a bounded one-environment prototype while keeping full task training gated. Native integration checks later passed in their scope, but physical pickup remained unresolved.

5. The first knife-grasping campaign

5.1 Why touching the handle was not enough

The early central-pickup tests used a free knife lying on the board. There was no attachment or in-episode knife teleport. The arm approached through joint drives and physical contact. Initial opposing pinch contact was achieved, but the knife did not remain securely carried during clearance. [S03]

The long knife also constrained placement: the recorded knife length was approximately 43.8 cm on a board approximately 46.9 by 29.9 cm. A grasp that looks reachable may still place the blade outside the safe board footprint.

5.2 Tilting created access, not an unsupported pickup

The team tried lifting the handle while the blade remained supported on the board. Early tests obtained about 15 mm handle rise, followed by a supported plateau. Later palm-seating work raised the handle to about 36.8 mm.

This was real contact-mediated motion, but the board was still carrying part of the load. It did not meet the unsupported hold requirement. The distinction prevented a visually impressive tilt from being reported as a completed grasp.

5.3 Seating the handle in the palm

The next idea was to preserve the thumb-index pinch while moving the wrist and other fingers toward a more secure wrap. Offline paths constrained selected contact points and normals, while the runtime controller checked tracking and drift.

A small final adjustment preserved the pinch with about 0.807 mm additional lateral motion. It did not establish palm contact or a three-finger wrap, so the system refused to lift. In other trials, contact vanished during motion before a large force spike. The cause was not conclusively reduced to one friction or collision issue.

5.4 Transferring load between fingers

Six bounded contact-transfer trials explored acquisition angles, ring clearance, middle-finger staging and rolling. Stable transfer to the additional fingers was not demonstrated. Other launches were refused by the memory reserve instead of proceeding under unsafe resource conditions.

The important geometric lesson was that curling fingers more does not always bring their useful inner surfaces closer to the handle. Depending on wrist and handle orientation, more flexion can move the contact surface away or push the knife against the board.

5.5 Offline grip checks changed the diagnosis

Offline geometry and static load checks found that some proposed layouts lacked useful opposing inner contacts or could not balance the knife's turning moment. A rolled, approximately edge-on terminal proposal had conditional static support under the assumed loads. [S03]

Static support is like balancing a wrench on paper: it tests whether available contact forces could resist weight and torque. It does not prove that the robot can reach that grip, establish it dynamically, lift without collision, and keep it during cutting. This distinction remains relevant to the November pickup work.

6. The first custom tomato model

6.1 Why a separate mechanics model was explored

The capability audit looked at installed source rather than assuming an engine could perform the required cut. Newton offered particle, spring and contact building blocks, but arbitrary fracture and supported coupling were not automatic. Coincident spring endpoints were also unsafe in a source path that divided by separation. [S04]

The initial design therefore became a separate, inspectable mechanics model. It used two prepared sides of a potential end-cap cut, linked by damageable interface elements. That made one cut easier to define, but it restricted the cut geometry in advance.

6.2 Nodes, springs and interface patches

Nodes are points carrying the model's motion. Springs resist changes between connected points. The interface was grouped into patches, each with a saved area and shared damage history. The original registry had 47 patches and 403 interface bonds. [S04]

The physical intent was that intact material transmits force, damage reduces its resistance, and fully broken connections no longer provide cohesion. A patch reaching complete failure is not the same as the entire tomato becoming disconnected: other patches may still bridge the two sides.

6.3 The first fracture-energy failure

The first scalar-to-bundle damage mapping did not spend the intended work. Opening energy was about 13.60% too high and shear energy about 64.20% too low. A scalar law that works for one ideal spring did not automatically work for a bundle of oblique finite-length springs.

The correction used the actual stored patch energy when evolving damage. Focused tests passed, and all 47 real interface patches subsequently passed opening and positive/negative shear checks: 235 cases under the unchanged 1% criterion. This qualified that law in those fixtures, not real tomato tissue.

6.4 Closed cut surfaces and sensible mass

Separate closed surfaces were created for the slice and remainder, with matched but independent cut-face nodes. Matching positions are useful for an initially closed interface; separate identities allow the faces to open later.

A provisional fixed surface-mass allocation was replaced by volume-based nodal masses. Directional spring stiffnesses were refined and small-strain compression/shear tests passed. These changes improved the model's internal consistency while leaving it explicitly synthetic and uncalibrated.

6.5 The large-deformation failure

Small-strain tests were not enough. Larger compression and shear inverted some local volume cells, even when the equilibrium force residual and some stiffness checks looked acceptable. An inverted cell means the local material mapping has folded through itself.

The repair added local volume energy to resist collapse. The model was now a hybrid spring-and-volume model, not a pure spring network. It passed 108 protected load stages across 18 compression/shear paths in the recorded study. This is why merely watching the outside of the tomato was insufficient.

6.6 Mesh refinement was its own investigation

Changing node spacing changed the response. A coarse mesh failed a selected comparison, and the first 4 mm geometry had duplicate-node and local-inversion issues. Reusing same-piece nodes and refining the remainder repaired those defects.

The slice still needed more internal sampling through its thickness. After that change, the refined slice family met the recorded 10% comparison target across the tested directions. These were limited mesh-sensitivity results, not universal convergence or calibration.

7. Contact, damage and the slow insertion campaign

7.1 Separating cohesion from compression

Cohesion holds two cut faces together while they are intact or partially damaged. Contact prevents them passing through one another when they close. A broken face should lose cohesion but should still collide with the other face.

An interface correction moved compression responsibility into unilateral contact. Unilateral means it pushes when surfaces try to overlap but does not attract separated surfaces. This avoided using tensile cohesion as an artificial compressive glue.

7.2 A first partial insertion

A surrogate wedge was coupled to irreversible damage in a shallow offline test. At 4 mm insertion, two patches had begun to damage but none had failed completely. The model then passed scoped unloading and supported-reclosure checks without healing the saved damage. [S04]

These solves were quasistatic: each step sought a force-balanced state rather than simulating all fast dynamics. They were useful for studying contact and work, but they were not a native robot cutting episode.

7.3 The between-vertex penetration defect

Checking only material vertices missed a blade crossing through the interior of a surface triangle. The vertices could all appear clear while the blade intersected the face between them. A finite-triangle counterexample stopped the deeper full-cut attempt.

The contact backend was corrected to reason about finite triangles and weighted forces. This is a general numerical lesson: sampling a few points does not certify an entire surface or volume.

7.4 Why a converged endpoint was not enough

The system also checked the swept path between accepted endpoints. A blade and material could be nonpenetrating at both ends while intersecting in the middle. Tight swept-clearance requirements forced smaller insertion increments near difficult contact changes.

Smoother cubic candidate paths reduced some measured penetration but did not rescue every interval. Rejected endpoints and trajectories stayed rejected. A nicer curve was not treated as permission to loosen physical acceptance limits.

7.5 Fixing the numerical search

The nonlinear solver had to balance forces while contacts activated and released. One search failure involved contacts predicting attractive loads that should have been released. Another involved a fallback that discarded useful off-diagonal stiffness information.

Separate search corrections preserved the physical energy and tangent while improving how candidate states were found. Independent audits still decided acceptance. This distinction matters: a numerical search aid is not a change to the material law, and an optimizer's approximate curvature is not a physical stability certificate.

7.6 What 8.84375 mm meant

One accepted branch reached 8.84375 mm insertion with two fully failed patches and one connected material component. Its withdrawal and supported reclosure were checked from that actual history. Damage was preserved, and the blade returned to zero insertion without pretending the remaining connections had disappeared. [S04]

The withdrawal branch and later forward-insertion branch must not be spliced into a fictional single successful cut. Each result belongs to its own saved predecessor chain.

7.7 What 9.638916015625 mm meant

The preserved October 27 forward checkpoint reached 9.638916015625 mm. Five of the 47 patches were fully failed; 42 still provided active connection, and the graph remained one component. It was progress in a difficult numerical continuation, not complete cutting or release. [S05]

An old dashboard sometimes displayed approximately 8.844 mm. That was a historical branch view, not the latest authoritative forward state. This is one reason later summaries needed source-bound evidence rather than reading a single visual progress label.

8. The bounded offline preparation period

8.1 Why the priority changed

The October 27 approved 15-hour preparation plan superseded further tiny-depth continuation. It redirected effort toward inventory, backend limits, units, history, resets, work accounting, guarded workers, mechanics evaluation and learning-interface preparation. [S05]

The aim was to make the next runtime stage concrete and testable, rather than spending the entire window advancing the knife by tiny fractions of a millimetre.

8.2 Contracts that prevented subtle mistakes

SI contracts fixed metres, seconds, newtons and joules. Frame contracts specified where forces and moments were expressed. Time contracts distinguished instantaneous force from accumulated impulse. History contracts prevented rejected trial steps from permanently damaging material.

Reset tests checked that episode history was cleared. Observation masks separated quantities a future robot could measure from privileged simulator information. Work ledgers separated stored energy, dissipated fracture work and external input.

8.3 The native-worker route was initially disabled

Separate worker/configuration code was prepared for small contact and two-piece tests, with execution guarded. Source inspection identified unresolved native integration and coupling issues. Static checks and process fixtures were not called native physics validation.

The preparation window ended on October 28. Quota-reading failures and an unfinished final handoff were preserved. The old automation was later deleted. Subsequent

November runtime work belonged to later authorizations; it was not a silent restart of the expired overnight window.

9. October 31 to November 2: from isolated mechanics to native coupling

9.1 Preserving the work in Git

The project workspace was not itself the publishing Git checkout. A separate checkout was prepared to archive code, plans and evidence into TianjinUniversity. Large assets were left local and listed in a manifest. This protected the original working directories and other contributors' edits. [S18]

Authentication was initially successful as an account, but push access failed with HTTP 403. A token authenticates an identity; it does not grant repository rights that identity lacks. Later write access became available, and the snapshots reached the research branch.

9.2 Native contact and force-transfer tests

The next stage built increasingly demanding tests of geometry, force signs, reference frames, contact buffers and reaction transfer. A force applied at the wrong point can have the right magnitude but the wrong torque. An impulse divided by the wrong duration becomes the wrong force.

Small rigid probes, articulated examples and replay tests were used before claiming a mounted-robot result. Passing these tests established pieces of the coupling interface, not a complete cut.

9.3 Integrating the actual saved material

The November 1 report records the actual 4,032-node model with 33,996 bulk springs, 16,291 volume cells, 5,504 boundary triangles and 47 shared-history patches in the native implementation. It compared assembled forces and numerical derivatives against independent offline calculations. [S06]

Prescribed opening fixtures drove all patches while recording energy and support impulses. These were controlled material tests. The knife did not independently discover and cause every fracture event in those fixtures.

9.4 Keeping the original blade geometry

The saved blade had 74 vertices and 144 triangles. A constrained decomposition produced solid cells retaining that exterior, rather than replacing it with a larger convex hull. Surface triangles could then be clipped against blade volume to find intersections that vertex-only checks miss.

An earlier unconstrained decomposition failed and was rejected. The distinction matters because a simplified collision hull can change the effective sharpness and contact location even when the rendered knife looks unchanged.

9.5 The fixed mount was a deliberate simplification

The robot scene used an ideal fixed knife attachment in this branch. That allowed the team to investigate arm motion, material coupling and cutting without requiring a contact-only grasp to succeed first. The mount was disclosed; it was not evidence that the fingers could carry the knife by friction. [S07]

Later changes to stance, elbow placement, approach and withdrawal improved the visible route. A dry route can prove reach and clearance in that route, but it cannot prove loaded cutting or a stable free grasp.

9.6 The November 2 review corrected overbroad milestones

The review kept scoped component passes while reopening whole-system questions: sustained support, mixed material loading, mounted feedback, loaded reset, motor work and complete separation. It also kept experimental tomato validation pending. [S07]

A published force curve was considered a provisional reference. A plot alone cannot identify all material/contact parameters, and fitting points on one curve is not independent validation. The saved data-request draft was not sent.

10. November 3 to 7: visible robot motion and research cutting

10.1 Motion and grasp appearance developed alongside mechanics

The saved work includes kitchen stance, pickup demonstrations, overhand and wrapped-grip variants, and backward-elbow positioning. These changes made the sequence easier to view and execute in the scene. Their existence does not retroactively turn a fixed mount into a qualified finger-contact grasp.

This branch was useful because it separated scene and kinematic issues from material issues. A visible, reachable trajectory was a prerequisite for testing the cutting model under the actual arm.

10.2 Live fracture needed transactional history

The research slicing extension added a transaction mechanism: trial damage was computed separately, then committed only if the mechanical step passed. If the solve failed, the old state, work and damage were restored. [S08]

Prescribed patch opening agreed closely with the expected damage law, but loaded solves still failed. A decreasing energy value alone did not guarantee an acceptably

balanced nonlinear solution. More iterations did not automatically repair the problem.

10.3 Smaller steps helped one control, not the entire cut

A very small timestep repaired a particular loaded control. A separate long downward route was kinematically reachable. Those were different tests: one checked mechanics and the other checked reach.

Neither established complete cutting. Collision, support, contact and force/work qualification still had to hold together during the actual route. The later mounted continuation remained slow and incomplete in the stricter research model.

11. November 8 to 10: DiSECt and the search for a usable backend

11.1 Profiling showed where time went

A short current-solver profile took about 62 seconds to simulate roughly 0.00667 seconds. Nonlinear solving and contact geometry dominated. Many device-to-host array calls also synchronized the GPU. File recording was a small measured cost. [S09]

This finding mattered: removing the kitchen display or deleting evidence logs would not solve the main bottleneck. Slow execution was also not proof of physical accuracy.

11.2 Stock DiSECt looked faster, but was not an adoption pass

The official prism example advanced five simulated seconds in roughly ten to eleven wall seconds in the local comparison. It was a different specimen, mesh, contact model and motion. Native cutting failure flags and shutdown crashes were also recorded.

Therefore the result was promising feasibility evidence, not an equal-accuracy speedup or a ready tomato backend. The later modified candidate must not be described as though it were the untouched stock implementation.

11.3 Transferring the research interface exposed a coverage gap

When the original 47-patch law was mapped to new cut geometry, a negative control found that 303 of 604 interface vertices had no nonzero participation in the mapped quadrature. Moving an uncovered vertex could change neither cohesive force nor energy. [S10]

The approved repair used full-surface sampling while retaining the total fracture budget and original patch laws. All interface vertices became covered. The distribution changed, and that change was documented rather than disguised as a pure speed optimization.

11.4 Coupled settling and contact remained expensive

A 100 ms assembled run had finite material state but was not settled. Larger-step and implicit candidates encountered convergence or physical-stop failures. Some algebraic optimizations matched a single step yet failed to make the complete system practical.

Later Section 4 work reached a compression stop before fracture. A separate softer material candidate used a literature-based prior, not measured calibration. Its coarse hold passed its own guards; the saved November 10 status still showed the refined comparison pending. This history does not support calling the strict DiSECt-derived tomato completed.

11.5 CRESSim was tested, not simply chosen

The public CRESSim-MPM release was built in isolation and ran fast on small fixtures. The team then tested what happened after the blade left. Nearby intended pieces shared motion through the background grid. With a blade held in a gap they could appear independent; with the blade absent, unwanted transfer returned. [S11]

In plain language, the cut could depend on the knife still acting as a barrier. That is not persistent separation. Stock CRESSim was therefore not adopted unchanged. The test also did not provide calibrated tomato cutting forces or anatomical tissue behavior.

12. November 10 to 13: the custom numerical research sequence

12.1 Why so many versions existed

The v01-v27 series tested different numerical representations and repairs. It was not 27 successful tomato simulators. Some versions passed a small component and exposed the next failure; others were rejected immediately. Appendix B explains each recorded stage individually.

The recurring questions were: does the material remember cuts; does its motion agree with its strain; can thin fragments be represented; can contact cover the full volume; and can the complete loop run fast enough for learning?

12.2 Persistent history solved one problem

The first persistent-history prototype kept cut-side information after the blade withdrew. Restricted geometric fixtures showed that separated pieces could move independently and partial cuts retained bridges. This addressed the unwanted reconnection seen in the public CRESSim tests.

However, those cuts were geometry-driven fixtures. A physical-model fracture law still needed to pay for new surface, and blade and fragment contact still needed validation.

Bulk-only timing was already a concern.

12.3 Invisible deformation was a deeper failure

A later loaded test found that reconstructed geometry could fold while the stored strain scarcely changed. The force law was effectively blind to some motion. A volume guard detected this, but detecting a failure does not repair the formulation.

An APIC-style repair removed the reproduced invisible mode but introduced or exposed timestep-dependent damping. The next consistent-mechanics formulation kept position, strain and forces tied to one displacement field and avoided repeated transfer filtering. Intact mechanics then passed within its tested envelope.

12.4 Cuts require enough mathematical freedom

A displacement basis describes the shapes of motion the model can express. If both sides of a crack still share the same movement coefficients, the crack cannot actually open. If new coefficients have no properly integrated mass or stiffness, artificial free modes can appear.

Cut-aware integration fixed an off-grid sampling defect. Restricted multicut fixtures then produced independent pieces. Later local-continuity work repaired a different defect where a shallow partial crack had zero opening rank despite having a geometric interface.

12.5 Accuracy was not just a smaller timestep

Fine crack-front modes can move rapidly. A timestep stable for an intact body may be inaccurate after refinement. The sequence investigated timestep error, activation timing and changes to the representation itself.

A localized event scheme passed a restricted comparison, but at such small timesteps that the performance review rejected it as the laptop's online-learning engine. This was an engineering decision about that implementation and accuracy requirement, not proof that all MPM or cutting simulation is impossible.

12.6 Local mass and selective aggregation

A later branch used a more local representation and linear tetrahedral elements. It repaired an angled-cut mechanical defect but still suffered from very small cut regions, which can force extremely short stable timesteps.

Aggregation ties weakly supported numerical degrees of freedom to better-supported ones. Too much aggregation changed the response beyond the accepted comparison. Selective aggregation passed under an explicitly approved material-weighted velocity measure, while the original stricter pointwise failure stayed documented. That approval was material to the result.

12.7 Real software optimizations improved speed

Caching unchanged geometry, compiling convex operations, reusing GPU arrays and graphs, and overlapping independent preparation with stepping all produced measured improvements. These were meaningful implementation results.

By v22, repeated geometric diagnostic episodes exceeded about 100 aggregate transitions per second in the selected batch. But the benchmark still omitted physical blade contact, paid fracture, robot dynamics and policy inference. It had not yet met the complete-task learning requirement.

12.8 Contact brought the cost back

Adding original-CAD contact exposed sampling and nearest-triangle problems. Refined contact and independent comparisons repaired important component defects. A thin blade is especially sensitive to selecting the wrong nearby triangle or using insufficient numeric precision.

The cost remained high. A later shallow work-paid crack advanced successfully in a tiny coupon, but required repeated geometry and mechanics reconstruction. Optimizations improved exact searches and reduced redundant work without delivering the requested complete fast tomato.

12.9 The v27 decision point

Version 27 ended with the objective explicitly not achieved. It preserved substantial geometry, contact and rollback improvements but did not establish a full fast cut. The concrete choice became whether to continue strict unchanged-physics research or authorize an approximate training model. [S12]

This is the major fork in the project's history. The next model was not a claim that all the strict failures had somehow disappeared. It was a different, explicitly approximate scope.

13. November 13: the approximate model became usable

13.1 What changed in v28

The approximate model used 2,803 massive particles and 31,950 initial elastic bonds, with 3.5 mm spacing and about 121.253 g total mass. Blade motion spent irreversible work to break existing connections; a future fracture plane was not supplied in advance. [S13]

It kept partial bridges, tracked disconnected components after completion, and retained particle contact. The blade was a tapered proxy rather than the original CAD. The particle spheres occupied only about 38% of the former continuum volume, making the spaces

between them an explicit approximation.

13.2 What actually passed

The restricted pilot completed work-paid cutting and horizontal withdrawal in all eight benchmark bodies. A subsequent perpendicular cut from an already cut state produced four components, and a larger tomato case also completed. Reset, parallel-state isolation and several stopping conditions were tested.

The final benchmark measured about 136.8 complete task transitions per wall second across eight environments. It included task physics, actions, observations, rewards, termination and selective resets. It excluded articulated robot dynamics, rendering and policy inference. No policy was trained in that benchmark.

13.3 What the pass did not mean

The model's imposed-strain response differed substantially from the old continuum reference. Coarse and fine cutting cycles could finish with the same broken bonds while differing by about 46% in peak force. An oblique smaller-tomato case hit the force stop and was not qualified.

The result justified a restricted approximate-task learning trial. It did not justify claiming realistic force prediction, arbitrary-path robustness, anatomical skin/pulp/seed mechanics, or a complete robot-learning throughput figure.

13.4 Why this is useful anyway

A deliberately approximate environment can be useful for early control experiments if its limits are visible and the controller is later tested against stronger references and reality. It lets the team investigate task interfaces and behavior without waiting indefinitely for a research-grade custom continuum solver.

The risk is learning an exploit of the approximation. That is why future evaluation must include held-out cases, force behavior, stopping rules and external validation rather than accepting a high reward alone.

14. Appearance, speed of motion and the integrated robot

14.1 Smooth appearance was a rendering upgrade

Version 29 added a smooth tomato exterior, stem, calyx and cosmetic flesh/seed colors. The display followed material motion and broken local connections, rather than swapping in a preset sliced tomato. [S14]

The smooth surface visually filled particle gaps; it did not fill them with physical material. Cosmetic seeds were not simulated seeds. The raw particle view remained available so

appearance would not conceal the underlying collision representation.

14.2 Faster cutting motion was tested separately

Version 30 changed the controller while preserving the approximate material model. Doubling and quadrupling the feed reduced simulated episode duration in selected cases, but could increase load. A smooth motion variant failed when continuous feed coincided with sliding reversals. [S15]

A synchronized variation improved that particular behavior. These were scripted comparisons, not learned optimal cutting. Timestep sensitivity remained significant, and failed partial runs could not be compared to completed cuts as though their shorter time meant success.

14.3 Version 31 completed a native robot cycle

The approximate material was connected to measured native blade motion in the kitchen scene. The material reaction force and moment were returned once to the native tool, with a declared one-millisecond delay. This was more than moving a rendered blade through a saved animation. [S16]

The successful headless cycle broke 1,857 bonds and produced two components, with about 46% of the mass in the smaller piece. It completed horizontal withdrawal and settling. The saved report records a peak summed simulated load of about 1.3505 N and particle penetration of about 0.0081 mm.

14.4 Important simplifications remained

The knife was held through an ideal fixed attachment after the hand closed. The tomato remained supported through its bottom layer. An ideal retracting cradle cleared the route after pickup. The original tool mass/inertia were retained as a disclosed approximation despite the fitted shorter blade.

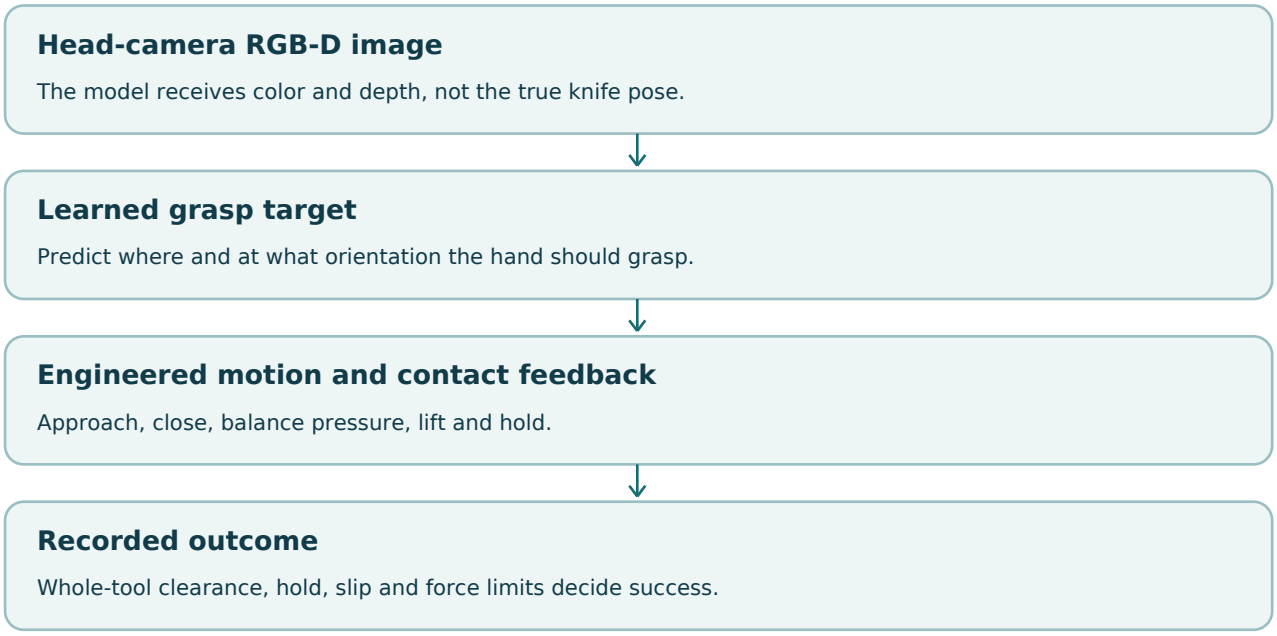
Therefore this was not a demonstrated free-finger grasp carrying cutting loads, nor full dual-arm tomato holding. It was an operational coupled demonstration with clearly described support assumptions.

14.5 The failed integration runs mattered

An early run cut and withdrew but did not settle quickly enough, so its overall result remained failed. A longer bounded settling rule later met the same quiet-state condition. Another run hit a board-contact problem; the contact envelope was corrected while retaining geometric clearance and physical stops.

The successful headless entry/cut/withdrawal/settling section took about 366 wall seconds. A visible GUI cycle also completed but took longer. Neither number was an end-to-end high-throughput robot-learning benchmark. Robot loaded reset and motor-work qualification remained limited.

15. November 13 to 14: learning where to grasp the knife



15.1 The chosen hybrid approach

An initial fully learned image-to-joint approach failed live pickup tests. The user chose a more staged approach: learn the visual grasp position and orientation first, then use a tested motion controller to execute the pickup. Full-motion learning could come later. [S17]

This is a hybrid learned-and-engineered system. The camera model chooses a target; the controller approaches, closes, balances contact and lifts. Calling the entire motion a learned policy would overstate what happened.

15.2 Finger contact replaced the attachment in this separate branch

Unlike the fixed-mount cutting demonstration, these pickup trials used a free knife supported by finger contact and friction. Successful trials had to lift the whole knife clear and hold it unsupported for three seconds.

Closing fingers in the image was not sufficient. Blade clearance, slip, contact load and sustained holding all mattered. This made the result more meaningful than a visually plausible grasp pose.

15.3 The first completed evaluation

The frozen first version passed all three validation placements and seven of eight fresh test placements on the original short-knife setup. One placement lost its grip and remained a failure. A blank-image check rejected the estimate before approach rather than using hidden simulator position. [S17]

That is a useful measured result, but eight tests on one limited setup are not broad statistical evidence for every knife or board position. The two untrained shape probes and opposite-face probe initially failed.

15.4 Why more force was not the answer

Increasing holding effort worsened one slipping case. Recorded contacts showed the fingers twisting the handle against the thumb. Later changes waited for opposing contact, balanced pressure while the board still supported the knife, and used a bounded readiness period before lift.

A grasp must resist a turning moment as well as weight. Pressing harder at the wrong place can rotate the handle out or wedge it against the board. This is why contact location and closing sequence became more important than simply stronger commands.

15.5 Expanding the dataset and cases

The next isolated version covered three knife shapes and both resting faces, with a wider placement range. It collected 240 training images and 12 separate validation images. Fourteen final evaluation cases and an unseen knife variant were reserved. [S19]

Some development grasps were first tested using supplied target positions. Those proved physical feasibility for the controller at those targets, not camera generalization. Five of six shape/face combinations passed such development checks; the heavier knife on its opposite face remained problematic.

15.6 Vision accuracy improved, but grip remained the bottleneck

One expanded model reduced worst validation position error from about 7.9 mm to about 1.4 mm. Later target revisions and a visual selector produced their own measurements; the latest inspected chat reported 1.37 mm maximum error with all 12 saved-image estimates accepted.

These numbers belong to different model/target versions and must not be pooled as one result. Even a one-millimetre placement difference can change which finger surface touches a sloping handle. Accurate image predictions do not guarantee stable contact.

15.7 The latest unresolved experiments

The newer camera-driven comparison produced several successful pickups, but other cases hit force limits, slipped or encountered a blank startup frame. A camera-readiness correction fixed the blank-frame problem without resolving grip retention.

The latest inspected chat reports that finer timesteps and another constraint solver both still encountered force spikes on the heavier flipped knife. It then returned to grasp geometry and began testing load sharing across whichever opposing fingers actually touched, rather than requiring the index finger alone. This was ongoing work, not a completed acceptance result. [S19]

15.8 Why these tests are kept separate from cutting

The pickup prototype has not been shown here to retain the knife under the full cutting load of the integrated tomato demonstration. The integrated cutting demo used a fixed mount. Combining their successes on paper would hide an untested interface between them.

The next bridge is therefore explicit: qualify the free grasp under relevant tool loads, then evaluate the coupled task. That is a future gate, not something a successful unsupported hold already proves.

16. What has actually learned, and what has not

16.1 Resource PPO versus task learning

The October PPO updates were a resource and software exercise with a posture reward. They did not learn pickup or cutting. The November camera model learned visual target prediction. The approximate tomato benchmark supplied an environment suitable for an initial restricted trial but did not train a policy.

These three activities can all contain machine-learning software while having very different meanings. Keeping them separate prevents the phrase 'training worked' from hiding which skill was actually evaluated.

16.2 Observations, actions, rewards and episodes

An observation is what the controller receives, such as an image, tool pose or force signal. An action is its command. A reward is a training score. An episode is one attempt, including a defined reset and termination.

A useful reward cannot compensate for invalid physics. If material reconnects after cutting, a learner can exploit the mistake. If reset leaves damage behind, later episodes start with easier tomatoes. That is why history, masks and resets were developed before claiming learning readiness.

16.3 Privileged information must be disclosed

A simulator knows the exact knife pose and internal damage state. A real camera-equipped robot may not. Such information can be used for training labels or diagnosis, but should not silently enter the evaluated image-only controller.

The newer pickup path explicitly separates teacher-target development checks from camera-driven runs and rejects missing images. This is a methodological improvement even while generalization remains unresolved.

17. Publication, preservation and the paper direction

17.1 The November 14 repository update

The research branch codex/tomato-preparation-20261031 received the newer code, reports, successful and failed evidence, and dashboard snapshots. The main update changed 29,344 files. A follow-up clarified which small vendored source snapshots remained. The verified remote tip was 5f7267f358e8c689ccaa300bd6824ab19436aa2c. [S18]

Large assets, arrays and model weights remained local and were indexed. The original source workspaces and ongoing experiments were not changed by publication. The branch was not merged into main in this work.

17.2 A snapshot is not a portable finished product

Some files are live progress records copied while work is ongoing. Each included file can be hash-verified without the entire collection representing one atomic experiment checkpoint. Machine-specific paths and omitted dependencies still need attention before reproduction.

The repository is valuable as a research archive. It should not be presented as a one-command, calibrated, fully trained system merely because it contains many files.

17.3 How the possible paper contributions narrowed

The discussion considered persistent fracture, arbitrary paths, fast simulation, force-aware cutting, dual-arm coordination, robustness, realistic sensing, sim-to-real transfer and an evaluation framework. These were possible directions, not nine established contributions.

The provisional recommendation was one main contribution: force-aware adaptive cutting. Two supporting directions were robustness across tomato properties and control using realistic observations. The newer pickup work strengthens the perception/contact-control groundwork, but does not itself prove adaptive cutting

superiority.

17.4 The research claim still needs experiments

A defensible paper would compare a learned or adaptive controller against a fixed-motion baseline and a well-tuned conventional feedback baseline, then test held-out conditions and sensor ablations. The repository's research rules call for repeated seeds and evidence-backed claims.

If the team instead wants a simulation-method paper, the numerical formulation and its accuracy/performance evidence become the main contribution. That is a different paper scope. The project should not promise both a new high-fidelity simulator and a fully learned dual-arm system without deciding which evidence it can complete.

18. The current boundary and the next logical steps

18.1 What is demonstrated

There are scoped offline material/contact results, native coupling diagnostics, an approximate work-paid cutting task, a fixed-mount robot cutting demonstration, and limited successful camera-guided free-knife pickups. There is also substantial failure evidence explaining why stronger claims are not yet justified.

18.2 What remains unresolved

Real tomato material calibration, high-fidelity complete-cut convergence, general arbitrary-path reliability, fast full-robot training, robust free grasp across knife variants, free-grasp retention during cutting, active dual-arm coordination, motor-work qualification and real-world transfer are not established by this record.

Some components have exact resets; others restart a process or remain only partially tested. Reset claims must therefore name the component rather than treating every branch as one completed environment.

18.3 The practical sequence from here

First, finish the current pickup qualification without mixing teacher-target and camera-driven results or combining different controllers' success counts. Then decide how that grip will be tested under cutting loads. Keep the fixed-mount demo as a separate reference.

For learning, choose a restricted approximate task and declare its limitations. Measure the full loop being trained, compare against conventional control, and reserve held-out cases. For physical realism, obtain appropriate measurements and separate fitting data from validation data.

For writing, begin the system description, decision history and limitations now. Delay strong outcome claims until the matching experiments exist. A clear paper can explain a focused contribution without pretending that every branch of this long project is solved.

18.4 The central lesson of the project so far

The project did not advance along one straight line from a robot scene to a trained cutter. It repeatedly separated things that initially looked equivalent: touching versus holding, damage versus separation, geometry versus mechanics, appearance versus collision, fast kernels versus fast episodes, and simulation success versus real-world validity.

Understanding those distinctions is the most useful way to understand the work. They explain both why progress took many versions and why the later approximate cutting and vision-pickup results should be valued without overstating them.

19. A guided tour inside the simulation

19.1 What this expanded part adds

The first part tells the project's chronological story. This part opens the machinery behind that story. It explains what the model stores, what each calculation is trying to accomplish, and why results that look similar on screen can mean very different things physically. It retains the November 14 historical cutoff; it is not a new qualification campaign.

The numerical examples marked 'illustration' are teaching examples, not project measurements. Recorded values are tied to the source labels and version guide. The diagrams are explanatory schematics, not screenshots, scale drawings or simulated trajectories.

19.2 There is no single object called 'the simulation'

Think of the system as several cooperating descriptions. The robot has joints, rigid links, actuators and collision shapes. The tomato has material state, connections and a rule for resisting deformation. The knife has a pose, a visible surface and a contact representation. A controller decides what to do next. A renderer draws what the user sees.

These descriptions need not be solved by the same engine. In parts of this project, native robot dynamics and custom tomato mechanics exchanged information. That made it possible to keep the robot scene while changing the material implementation, but it also created an interface whose clocks, forces and coordinate systems needed checking.

19.3 Geometry says where; mechanics says how it responds

Geometry tells us the current shape and where surfaces meet. Mechanics tells us what force follows from deformation and how the body moves. A red shape can look like a tomato while having the mechanics of a spring network, a rigid sphere or nothing at all.

This is why the later smooth surface did not establish skin stiffness. The model would need a separate skin material law, thickness, mass distribution and coupling to the interior to make that claim. Drawing a skin color adds none of those automatically. [S14]

19.4 State is the information that survives between steps

A state may contain positions, velocities, material strain, damage, broken connections, accumulated work, support state and the simulation clock. Some implementations also store grid coefficients or contact-related caches. Different branches use different state definitions.

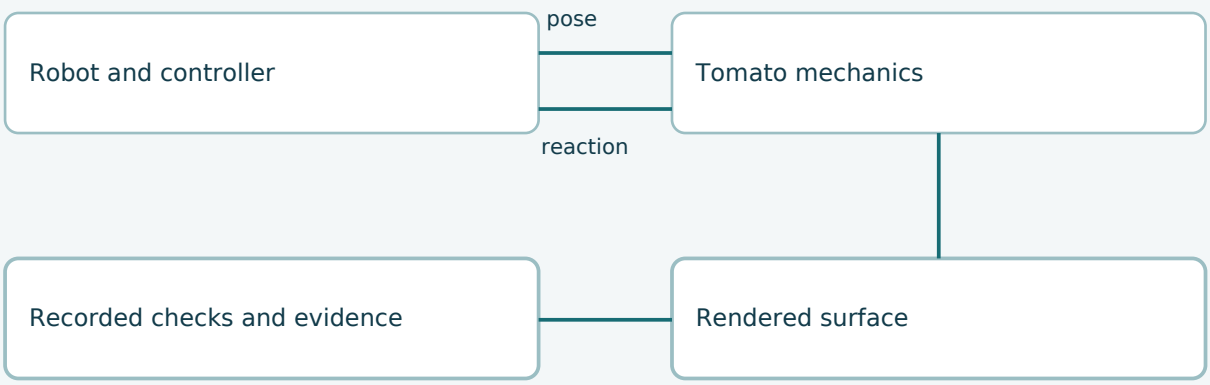
If a system saves only positions, it may restart with the right picture but the wrong physics. A damaged interface could become intact again. Momentum could disappear. A previously paid fracture event could be charged twice. The snapshot and reset work was aimed at these invisible differences.

19.5 Four questions to ask about any demonstration

First: what is being commanded directly? Second: what is actually moving because of calculated forces? Third: which supports or attachments make the task easier? Fourth: which success test was evaluated?

For the fixed-mount demonstration, robot blade motion drove the material and reaction returned to the robot, but an ideal attachment retained the knife. For visual pickup, the knife was free and retained by contact, but that branch had not completed the combined cutting task. These demonstrations answer different questions.

EXPLANATORY SCHEMATIC — not an experimental result



A convincing picture does not establish a correct force or cut.

20. The original spring-and-volume tomato, from the inside

20.1 Starting with a finite set of moving points

A real tomato has continuously distributed tissue. A computer instead stores a finite set of values. In the original model, nodes described material motion, springs resisted relative movement, and local volume cells helped detect and resist invalid collapse. Later native integration used 4,032 nodes, but the early geometry and refinements had their own counts. [S04, S06]

A node is not necessarily a literal grain of tomato. It is a numerical place where motion is represented. The spacing between nodes determines which local details can be resolved. A narrow feature between nodes may require additional geometric or integration treatment.

20.2 What a spring can and cannot do

For a simple teaching spring, force magnitude is stiffness multiplied by extension. Stretch it farther and it pulls harder. The stored energy grows with the square of extension. Actual project laws can be more complicated, but this basic picture explains why force and energy must agree.

One spring mainly resists motion along its own direction. A network needs enough useful directions to resist shear and changes of shape. A few plausible-looking connections do not guarantee that the assembled body has the intended stiffness in every direction.

20.3 Why masses matter even when the shape looks right

The same shape with twice the mass responds differently to the same force. Mass also influences vibration, momentum, settling and stable timestep. Allocating mass to surface nodes by an arbitrary percentage can produce an uneven dynamic model.

The project replaced an early surface allocation with volume-based contributions. This is a consistency improvement: represent how much material each node supports instead of assigning mass primarily for convenience. It is not proof that the resulting synthetic material matches a measured tomato.

20.4 Why local volume cells were necessary

Imagine a tetrahedron: four points forming a tiny pyramid. Its signed volume tells us both how much space it occupies and whether its orientation has flipped. A nearly zero volume indicates collapse; negative signed volume indicates inversion relative to the original orientation.

The original spring network could reach a force-balanced configuration while some volume cells inverted. Springs can find a low-residual arrangement that is physically unacceptable. A separate volume term added resistance as cells approached collapse. This changed the model into a hybrid, explicitly documented in the reports.

20.5 Why a volume guard and a volume force are different

A guard asks, 'Is this candidate state allowed?' A force changes the motion so the material resists reaching a bad state. A guard alone can stop an inversion but cannot tell the solver how to avoid it. A barrier-like energy supplies a resistance that grows near an undesirable volume.

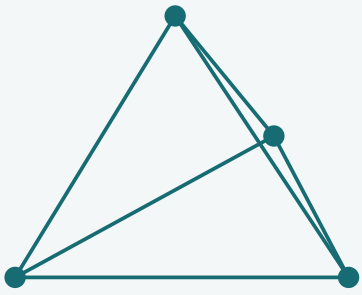
Both still need numerical care. A trial update can jump across the valid region unless the solver limits the step. The native implementation therefore used feasible-step reasoning as well as a final admissibility check. Passing the final volume check does not mean every arbitrary search update would be safe.

20.6 The prepared end-cap interface

The early geometry contained a selected end-cap plane, with a slice side and a remainder side. Their matching cut surfaces had independent node identities and were held together through cohesive patches. This makes a particular separation topology explicit and inspectable.

It also means the model was not initially a tomato that could be cut anywhere in any direction. The intended seam was prepared. Calling it a general arbitrary-cut model would ignore that design choice. Later branches explored ways to create new discontinuities from blade motion instead.

EXPLANATORY SCHEMATIC — not an experimental result



Nodes, connections and volume



Opening + shear + remembered damage

21. Cohesive patches: how an intact seam becomes damaged

21.1 A patch is a small piece of interface

The original interface had 47 patches. Each patch represented an area over which the two sides could transmit traction, meaning force per area. Before damage, opening or shearing the interface produced resistance. As damage grew, the cohesive contribution weakened. [S04]

This is easier to understand as many small adhesive regions joining two surfaces. The analogy is limited: tomato tissue is not literally glue. It explains the bookkeeping of a cohesive model, not the biological structure.

21.2 Opening, shear and compression are different

Opening pulls the faces apart in the normal direction. Shear slides them relative to each other along the interface. Compression pushes them together. These motions can have different mechanical rules and different contact consequences.

A completely failed interface should not keep pulling the faces together. But if the faces later collide, they must still push against one another. That compressive collision belongs to contact. Removing cohesion must not remove all collision response.

21.3 Damage is a memory, not a temporary displacement

Suppose an interface opens, becomes damaged, and then closes when the knife withdraws. Its current gap may be zero, but its history is different from an untouched interface. A no-healing law preserves that damage instead of restoring its original cohesive strength.

This is why the withdrawal and reclosure studies checked every saved damage record. The visual faces can look closed again while remaining partially broken mechanically. 'The crack closed' and 'the tissue healed' are not equivalent statements.

21.4 Fracture energy is an area-based budget

A common model assigns a fracture energy Gc, measured in joules per square metre. Creating more failed area should spend more irreversible work. If a patch's area doubles, its full-failure energy budget should scale appropriately under that law.

Illustration only: if Gc were 80 J/m2 and new area were 10 mm2, the budget would be 0.0008 J. The area conversion matters: 10 mm2 is 0.00001 m2. This calculation explains units; it is not a measured tomato value or a claim that every project branch uses the same law.

21.5 Why the first bundle mapping failed

A damage formula based only on geometric opening/shear was applied to a bundle of finite-length oblique springs. The bundle's actual stored energy did not match the assumed scalar reference. Consequently, opening spent too much and shear too little relative to the intended budget.

The correction tied the common patch damage evolution to the actual patch energy. The same test was then applied across all actual patches and multiple shear directions. This prevented a fix that worked only for one conveniently oriented ideal patch.

21.6 What patch tests still leave unanswered

A patch test can verify that a prescribed opening dissipates the intended energy. It does not prove that a moving knife delivers that opening, that the bulk deforms realistically, or that the resulting faces collide correctly after separation.

The complete task must connect blade contact, mechanical response, fracture eligibility and history. Many later difficulties arose at those connections rather than inside the isolated patch formula.

22. Contact: detecting the blade is not the same as handling it correctly

22.1 A distance query answers only one question

A signed-distance query estimates how far a point is from a surface and on which side it lies. A contact routine also needs a useful normal direction, the affected material region, and a way to distribute force and moment.

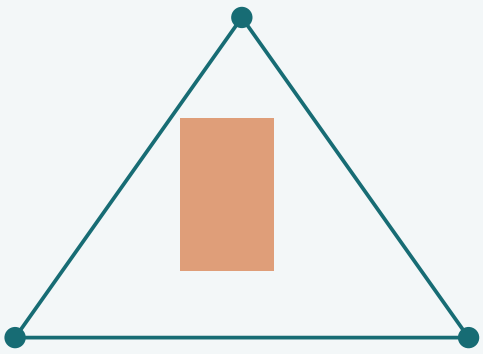
For a thin tapered blade, nearby triangles can have very different normals. Selecting the wrong triangle can produce an incorrect force direction even when the point is spatially close to the mesh. The later precision audit exposed this kind of issue. [S06; v23-v24]

22.2 Why checking vertices misses a crossing

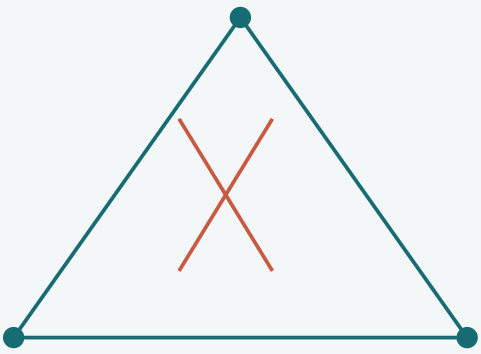
Picture a large material triangle and a narrow blade passing through its middle. None of the triangle's three vertices has to enter the blade. A vertex-only test therefore reports clear points while the triangle interior is intersected.

The offline counterexample showed that this was not merely hypothetical. The remedy considered the finite triangle and its intersection, rather than hoping denser endpoint checks would always catch it. A small set of samples is not a complete geometric certificate.

EXPLANATORY SCHEMATIC — not an experimental result



Clear vertices can miss an intersection



Check the finite faces and sweep

22.3 Why the whole movement between steps matters

Even perfect endpoint geometry is not enough for moving contact. A fast blade can start on one side and end on the other while both snapshots appear clear. The missing event happens during the interval.

Swept checks consider the intervening motion. They need an assumption about that path: for example, a piecewise-linear translation, a bounded rotation, or an interpolated material trajectory. A proof for one assumed path is not automatically valid for an arbitrary curved movement.

22.4 A contact force must reach the material properly

A pressure acting over a triangle is not automatically the same as a force at one vertex. The distribution should respect the interpolation of that surface. Otherwise the correct total force can still create the wrong local deformation or torque.

The project used weighted distributions and virtual-work checks to connect geometric contact to material degrees of freedom. These checks ask whether the force does the same work before and after changing representation. They are valuable because visually inspecting a push cannot reveal a subtle mismatch.

22.5 Why a convex hull may change the experiment

A convex hull fills concavities and can enlarge parts of the effective tool. For some rigid-object tasks that approximation is acceptable. For cutting, a change near the blade can alter penetration, local pressure and which material is eligible to fail.

The strict branch therefore invested in retaining the original CAD exterior. The approximate branch later used a declared tapered proxy. Neither choice should be hidden: one preserves geometry at greater numerical cost; the other changes the contact

model's scope.

22.6 Penetration limits belong to a representation

A 0.1 mm particle-knife penetration limit constrains the tested particle spheres. It does not certify every point in an imagined solid filling the spaces between them. A whole-volume bound on continuum cells addresses a stronger geometric question.

The same numerical limit written in two reports can therefore have different coverage. To interpret it, ask 'penetration of what, measured where, over which part of the timestep?' The approximate model explicitly did not inherit the old whole-continuum guarantee.

23. What happens during one cutting step

23.1 Start from an accepted state

The starting state is the last state that passed the model's acceptance checks. It includes material position and velocity, current connectivity and damage, work counters and clock information. A trial state is temporary until the complete update has been checked.

This approach prevents a solver iteration from causing permanent damage just because it briefly overshoots. The simulator can explore candidate positions without treating every numerical guess as a real event.

23.2 Read the actual blade movement

In a prescribed-blade fixture, the blade follows the fixture's declared trajectory. In the integrated robot, the material should receive measured native blade motion, because the real simulated arm may lag or deviate from the target.

Using only desired motion could let the tomato cut even while the robot had not reached it. The native integration's saved-pose replay was useful precisely because it checked material evolution driven by the actual recorded blade route. [S16]

23.3 Find contacts and eligible cutting regions

Broad-phase search first rejects material far from the tool. Narrow-phase calculations then determine nearby or intersecting regions more accurately. Cutting eligibility is a separate question: a blunt side contact should not necessarily create a new cut just because it pushes material.

The strict and approximate branches use different eligibility and fracture implementations. The shared intent is to avoid deleting material connections merely because a scripted progress variable says the knife should be deeper.

23.4 Compute material response and trial fracture

Elastic and volume terms resist deformation; contact adds external effects; the model updates velocity and position according to its integrator or solve. The fracture logic decides whether the declared loading and work permit damage or removal of a connection.

This is a conceptual order, not a claim that every implementation uses one identical loop. Implicit solvers can repeatedly solve these terms together, while explicit implementations can assemble forces and advance in smaller sequential stages.

23.5 Check and commit

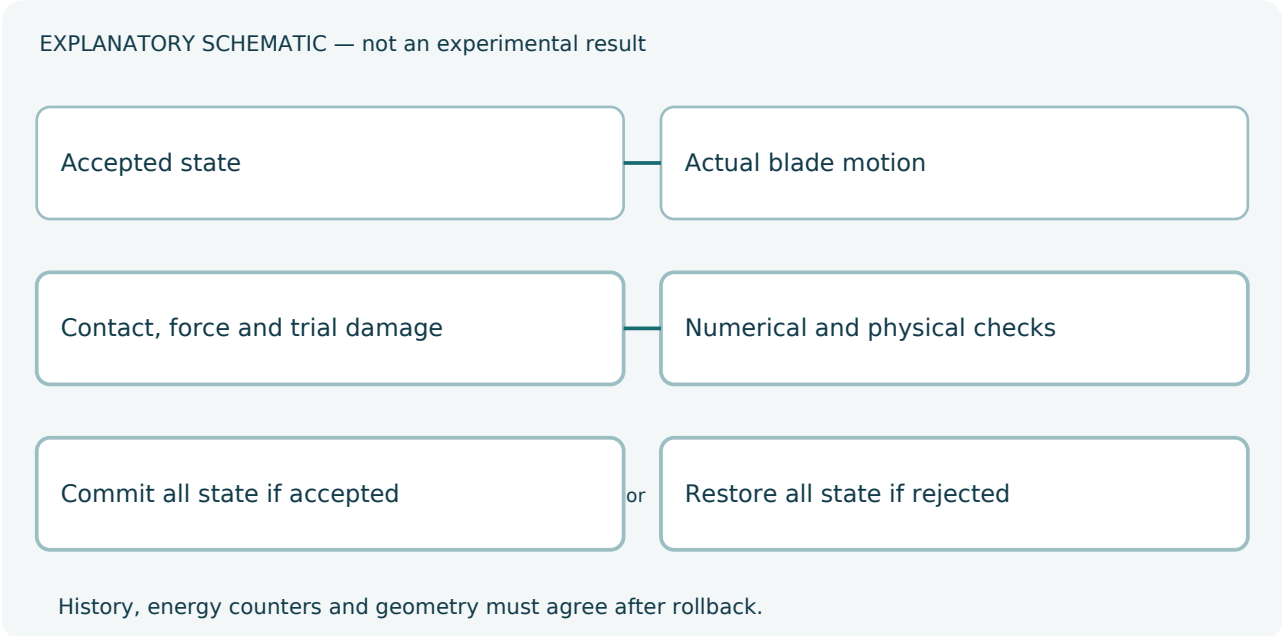
The trial may be rejected for excessive force, invalid material volume, excessive penetration, non-finite values, an unsupported topology or a failed numerical solve. On rejection, state and ledgers must return to the prior accepted values.

On acceptance, the state, damage and work become the new history. Repeated query calls must not charge work again or break more connections. This is why 'query' and 'commit' were separated in later transaction controls.

23.6 Return the reaction and continue

A contact force applied to the material should have an opposite reaction on the tool, with the corresponding moment about the chosen tool reference point. The robot engine then advances using that exchange according to the declared timing.

The one-millisecond delayed reaction in v31 is a numerical coupling choice. It is not identical to a single monolithic solve of every robot and material variable at once. Its adequacy requires coupled comparisons; an interface sign test alone cannot establish that.



24. Force, torque and energy: three different stories

24.1 A force sum is not always a sensor reading

Imagine two opposing local forces of 1 N. Their vector sum can be zero, while the sum of their magnitudes is 2 N. Both numbers are mathematically meaningful, but they answer different questions.

Several project reports distinguish a resultant wrench from a sum of local contact and cutting magnitudes. The approximate model's total-load stop must not be relabeled as a calibrated wrist-force sensor. The stock DiSECt comparison also showed why these quantities can differ greatly. [S09, S13, S16]

24.2 Moment depends on where the force acts

A small force far from the handle can create an important turning moment. To combine distributed contact into a tool wrench, each contribution needs its position relative to the chosen origin.

Changing from blade coordinates to world coordinates requires rotating both force and moment. Changing the point about which the moment is reported requires a lever-arm correction. A correct force magnitude with the wrong reference point can still twist the robot incorrectly.

24.3 Impulse must be divided by the correct duration

Impulse is the time accumulation of force. A contact API may report an impulse over a step rather than a force. If the step duration changes, dividing by an old hard-coded duration produces a wrong inferred force.

This explains the attention to clocks, integrated durations and source inspection. API names alone were not treated as sufficient evidence for physical units. A plotted channel called 'force' can still be wrong if its conversion is wrong.

24.4 Mechanical work tracks where energy goes

Force does work when its point of application moves in a compatible direction. That work can become elastic energy, kinetic energy, dissipated friction or fracture. Supports can also exchange work if they move.

An energy ledger compares these channels. A residual can reveal missing contributions, double-counted forces or numerical error. Small residual does not prove realistic material behavior: a wrong model can conserve its own wrong energy perfectly.

24.5 Illustration: a simple energy balance

Suppose an illustrative tool supplies 0.010 J. The body gains 0.003 J elastic energy and 0.001 J kinetic energy, while 0.004 J is assigned to fracture and 0.002 J to friction. Those channels balance exactly in this hypothetical example.

If fracture were accidentally counted twice, the balance would fail. If all numbers balanced but the fracture coefficient were ten times too small, the model could still cut much too easily. Numerical accounting and physical calibration are complementary, not substitutes.

24.6 Tool work is not yet motor work

The energy crossing the tool-material interface is not the complete energy used by the arm's motors. Motor work also involves robot inertia, gravity, drive behavior, transmission and other losses or modeling approximations.

The project explicitly kept motor-work qualification open. A passed material work ledger should therefore not be promoted into a claim of robot energy efficiency. A paper making that claim would need the appropriate joint-side measurements and accounting.

25. Why models can move in physically invisible ways

25.1 Position and strain must tell the same story

Position describes where material is. Strain describes how it has deformed relative to a reference configuration. If the code updates one without a compatible update of the other, the force law may not notice motion that should store elastic energy.

The v02 diagnosis found just such a mismatch in a particular transfer formulation. Reconstructed geometry approached or entered invalid volume while stored strain remained much closer to an undeformed state. The issue was structural, not simply a badly drawn mesh.

25.2 What a null mode means intuitively

Imagine a measuring device with a blind direction. The object can move along that direction without changing the device's reading. In a discrete material model, a non-rigid motion that produces no corresponding strain energy is a dangerous blind direction.

Rigid translation and rotation should not create elastic strain in an objective material model. Those legitimate motions must be distinguished from an extra deforming mode that is incorrectly free. A rank test counts whether the expected independent constraints or energy-bearing modes are present.

25.3 Why more iterations cannot fix missing mechanics

Iterations help solve the equations supplied to the solver. If those equations fail to represent a deformation mode, more accurately solving them does not create the missing resistance. It can produce a very precise answer to an inadequate discrete model.

Likewise, smaller timesteps can reduce integration error but cannot necessarily remove a structural null mode. This is why the project stopped to inspect operators and independent references rather than endlessly reducing the step after every failure.

25.4 Why APIC was tried

Particle/grid methods transfer information between material points and grid nodes. APIC carries local affine velocity information to improve how motion is represented during that exchange. The isolated repair tied position and constitutive strain to compatible grid motion.

It removed the reproduced invisible-mode fixture. However, the particular candidate still changed energy through repeated transfer, even without meaningful elapsed physical time in the diagnostic. That numerical filtering influenced timestep comparisons and motivated another representation.

25.5 The consistent-field repair

The v04 candidate stored generalized displacement and velocity on a fitted reference basis. Both material position and strain were reconstructed from that field. It used consistent mass and a time integrator without recurring transfer filtering. [v04]

The resulting intact controls passed at their selected small steps. The important achievement was not 'all physics solved' but removal of specific inconsistency and damping mechanisms in a restricted intact-body setting. Cut enrichment, contact and general shape had to be tested again.

26. Why a crack needs more than a line in the mesh

26.1 Shared movement prevents opening

If two sides of an interface use exactly the same movement variable, they cannot move apart independently. Drawing two triangles with separate colors does not change that shared degree of freedom.

A cut representation must introduce independent motion behind the crack while preserving continuous motion ahead of it. If it releases too little, the crack is locked. If it releases too much, the uncut bridge disappears before the knife reaches it.

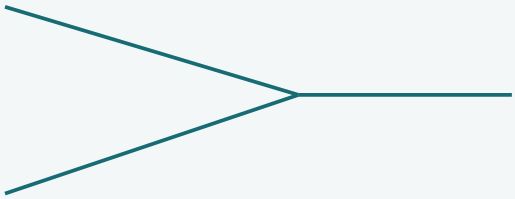
EXPLANATORY SCHEMATIC — not an experimental result

Tied crack lips



A damage label cannot create motion

Independent lips behind the tip



Opening behind; intact bridge ahead

26.2 What 'zero opening rank' meant in v25

The geometric interface existed and work had been charged for a shallow patch, but the displacement representation did not permit the two lips to separate. The opening operator had zero rank in that tested partial configuration: no available coefficient combination produced the needed relative opening.

A full-plane separation control could open. That contrast pointed to an overly restrictive local continuity rule rather than a universal inability to represent disconnected bodies. Version 26 changed the local field continuity and repeated the relevant tests.

26.3 Preserving state during a representation change

When a cut adds movement freedom, the existing deformed body should not jump back to rest. Its position, velocity and strain must be transferred into the new representation. That transfer can be exact for some compatible basis changes and approximate for others.

An exact transferred snapshot still does not guarantee identical future dynamics. New freedom changes the set of motions the system can take. This explains why the project checked both immediate transfer error and later trajectory response.

26.4 Why thin regions cause sampling trouble

Suppose a cut creates a thin sliver between old integration points. The geometry may recognize the sliver, but the old weighted samples may give its new motion mode no mass or energy. The mode then becomes unsupported or ill-conditioned.

Cut-aware integration divides the material into appropriate subregions and integrates those regions directly. The v06 repair addressed the tested off-grid defect this way. It did not simply add an arbitrary mass floor or delete the thin material.

26.5 A cut front is not an entire plane

A partial incision ends at a front. Material ahead of that front still connects the sides. A model that splits an entire plane at first contact can make cutting look easy by removing the very tissue the blade should still have to break.

The partial-bridge controls asked whether untouched material retained its connection. Later event-localization work asked when a local motion mode should be released as the blade's swept region accumulated. Those timing choices affect motion, not only topology labels.

26.6 Material history must travel with the material

If a piece moves or rotates after being cut, its cut history must move with it. A fixed world-space barrier can accidentally keep enforcing a cut at the old location or allow the moved pieces to share forces again.

Reference-coordinate and fragment-history approaches were explored to avoid that problem. Their restricted successes did not prove arbitrary curved cuts in freely deforming fragments, but they addressed why a world-fixed dividing plane is insufficient.

27. CRESSim, MPM and persistent separation

27.1 The background grid is a shared calculation space

In MPM, particles carry material information while a grid helps compute forces and motion. Nearby particles can contribute to the same grid nodes. That sharing is useful for intact material, but it must be handled carefully when two nearby groups are meant to be disconnected.

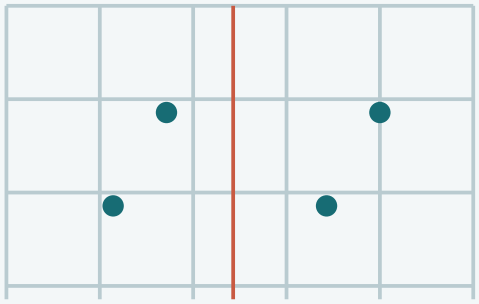
The key issue is not whether the renderer can show a gap. It is whether the calculation still mixes the pieces' momentum or constraints through shared grid channels.

27.2 Why the held-blade test looked better

The tested public CRESSim path filtered transfers using the current blade pose. While the blade occupied the gap, its side information could prevent sharing across it. Once the blade left, that temporary distinction was no longer sufficient. [S11]

The result was a model where nearby separated groups could influence one another again. Calling this 'healing' is an intuitive shorthand, but the measured issue was unwanted numerical coupling, not a verified biological healing law.

EXPLANATORY SCHEMATIC — not an experimental result



Shared grid can transmit across a cut

Fragment A: separate motion

Fragment B: separate motion

Test after the blade has withdrawn

27.3 The pulse test made the defect visible

One intended piece received a small velocity change. An otherwise identical baseline received none. The response of untouched pieces was compared between them. This isolates whether the supposedly disconnected parts share motion.

The report tested held-blade, absent-blade and widely separated controls. Their contrast was more informative than a single video of pieces appearing apart. It also showed alignment sensitivity, which cautioned against trusting one convenient grid-aligned example.

27.4 What persistent history adds

Persistent history retains information about separation after the tool leaves. Different fragment channels or cut-side histories can prevent formerly joined material from sharing the wrong quantities. The first custom prototype demonstrated restricted controls in this direction.

History alone does not supply physically realistic fracture. A model may remember a cut that was inserted geometrically without requiring any resistance or work. The next gate therefore connected topology changes to a declared physical-model law and contact.

27.5 Why a fast stock example did not settle the choice

The stock timing omitted robot dynamics, observations and the missing qualifications. It also used a different specimen, resolution and material setup from the existing research model. The numerical force behavior was not a measured tomato response.

The adoption decision was therefore conditional on function, not speed alone. A fast environment that allows an agent to exploit incorrect separation is not a sound shortcut to the intended research result.

28. Timestep, mesh size and numerical accuracy

28.1 Three resolutions that people often mix up

Spatial resolution controls how finely the body and contact are represented. The physics timestep controls how often dynamics are advanced. The control interval controls how often the controller chooses a new action.

They are related but different. At 50 Hz control, the action interval is 20 ms. A physics step of 0.2 ms means 100 physics substeps during that one action. Increasing control frequency is not the same as increasing material resolution.

28.2 Why stiff and tiny regions are difficult

A small mass connected through a stiff interaction can move rapidly. Numerically, its fastest time scale may be much shorter than the visible gross movement of the tomato. A small cut sliver can therefore limit the timestep for the entire explicit simulation.

This is why a slow-looking cut can still require extremely frequent numerical updates. The limiting frequency may come from a local mode near a cut front rather than the overall speed of the knife.

28.3 Stability versus accuracy

A stable run does not explode or grow without bound under its numerical conditions. An accurate run also follows a sufficiently consistent trajectory when resolution is refined. A stable algorithm can still damp motion excessively or shift the timing of force peaks.

The project repeatedly found that position or energy looked acceptable while velocity or force comparisons failed. Choosing whichever output happens to agree would hide the error rather than establish convergence.

28.4 Comparing at the same physical time

If two runs finish at different times, their final snapshots need not represent the same stage of a cut. A useful timestep comparison samples shared physical times or another carefully justified common event definition.

A coarse run and a fine run may break the same final bonds yet have different force peaks and material positions along the way. Same final topology is a useful discrete result, but it does not prove converged contact dynamics.

28.5 Mesh convergence is another experiment

A mesh comparison changes spatial representation, not just the integration step. The geometry, material parameters, loading and meaningful outputs should be controlled as closely as possible. Normalized energy or reaction can help interpret specimens with different sampling.

The early slice refinement repaired particular response differences. It did not prove that every future contact or fracture trajectory would remain within the same tolerance. New behavior can activate sensitivities that a static compression test never exercises.

28.6 Different error measures emphasize different risks

A maximum pointwise velocity error catches the worst location. A mass-weighted measure emphasizes how much material is affected. Neither is automatically universally best; they answer different scientific questions.

The selective-aggregation work passed after explicit approval of a mass-weighted contract, while larger local errors remained visible. Small mass does not necessarily mean unimportant contact: a tiny rapidly moving region at the blade can still matter. That limit belonged in the interpretation.

29. Solvers and why the GPU was not a magic solution

29.1 Explicit versus implicit advancement

An explicit method computes an update using available current-state information. It is often comparatively simple per step but may require a very small step for stiff interactions. An implicit method solves equations involving the next state, which can allow different stability behavior at the cost of nonlinear work.

Neither label guarantees speed or accuracy. An implicit method that repeatedly struggles to converge can be slower than many explicit steps. An explicit method with an enormous number of required substeps can also be impractical.

29.2 Nonlinear residual tells us how well equations are balanced

A residual measures the remaining mismatch in the equations being solved. A small force residual supports the claim that an equilibrium or implicit step has been solved to its declared tolerance.

It does not prove that the equations represent real tissue, that no cell inverted, or that time integration is accurate. This is why the project had separate residual, geometry, energy, contact and refinement checks rather than one universal 'passed' number.

29.3 A preconditioner changes the search, not ideally the answer

A preconditioner reorganizes a linear solve to make it easier. If used correctly to the same convergence criterion, it should not change the mathematical solution of that system. But improving one fixture's iteration count does not guarantee a whole coupled nonlinear trajectory will succeed.

The DiSECt-derived investigations contained such cases: a one-step equivalence or iteration improvement existed, while larger coupled steps still failed or the overall implementation remained slower. The report preserved that distinction.

29.4 GPU arithmetic can wait on CPU decisions

A GPU is efficient when many appropriate operations can run in parallel with limited overhead. If each small update requires copying arrays to the CPU, recomputing geometry in Python, and launching new GPU work, the device can spend substantial time waiting or synchronizing.

The profiler identified many host/device calls in the original coupled solver. Their time overlapped other measured phases, so it was incorrect to add every profile number as though each represented separate wall time. Profiling needed interpretation as well as collection.

29.5 Why caching and compilation helped

Geometry that has not changed need not be rebuilt identically every step. Reusing operators, compiling repeated geometric calculations and retaining GPU buffers reduced overhead. Matching controls checked that the optimized path still represented the same state and equations in the tested cases.

A cache is only correct while its assumptions remain true. A cut, deformation, changed selector or reset may invalidate it. The project preserved failures where a shortcut changed a tied selection or relied on stale data rather than accepting speed at any cost.

29.6 Why batching has limits

Several environments can share a GPU efficiently when their operations are compatible. Aggregate throughput counts transitions across all those environments, not the speed of one environment. Initialization, varied topology and reset can change the benefit.

Batching identical fixed states is useful for a microbenchmark but weaker evidence than independently evolving episodes. Later diagnostic loops added changing events and resets; the approximate benchmark then included a fuller task loop, still without the native robot.

30. Tiny cut regions, aggregation and the price of simplification

30.1 Why keeping every sliver can be expensive

A cut may pass extremely close to a node or element corner, leaving a tiny material region. Discarding it would change mass and geometry. Keeping it naively may produce a degree of freedom with very small inertia and a demanding dynamic time scale.

The project intentionally avoided silently deleting such regions or adding arbitrary mass just to obtain a larger step. Instead it investigated changes in numerical representation and then measured their effect on response.

30.2 What aggregation does

Aggregation expresses some weakly supported motion through better-supported nearby variables on the same fragment. Think of it as limiting an unreliable tiny numerical handle by tying it to a stronger representation of the material around it.

That explanation is conceptual. It is not a physical weld, and it must not tie separate fragments together. The actual reduced inertia and force mappings need to follow the declared transformation consistently.

30.3 Why the broad version failed

Version 18 improved a corner stability diagnostic but distorted the response compared with its reference. Even when both candidate and reference passed their own timestep refinements, their trajectories disagreed too much. [v18]

That finding ruled out blaming the entire discrepancy on a coarse step. The representation change itself mattered. An initial state projection with small error was not enough to validate later dynamics.

30.4 Why selective aggregation was different

Version 19 constrained only very weakly supported degrees of freedom, retaining more of the original motion space. It used the explicitly approved weighted velocity comparison and preserved local errors in the report.

The component result was useful, especially with a verified force-caching optimization. It did not mean all thin fragments, partial cracks or later contact were qualified. The mathematical approximation needed a clearly stated operating envelope.

30.5 Coalescence can accidentally close a crack

Later work explored merging compatible numerical descriptions to control growth in the number of coefficients. Matching the current positions and energies was not enough: two descriptions can coincide now but allow different future motions.

The correction required preserving the whole prior admissible field space, not just the saved snapshot. Otherwise merging could silently remove crack-opening freedom and effectively close a mechanically meaningful discontinuity. This is a subtle reason to inspect what a representation permits, not only what it currently displays. [v27]

31. The approximate v28 tomato explained carefully

31.1 This was an authorized model change

Version 28 did not claim to preserve the strict previous response within 5%. Its contract explicitly recorded that physics changed, while the production setup remained unchanged. It created a separate model for initial learning experiments. [S20]

This is not a minor implementation detail. It changes what conclusions can be drawn from successful cuts. The model can be useful as an approximate task environment without being a validated substitute for the previous continuum material.

31.2 The particle-bond representation

Particles occupy selected lattice locations inside an approximation to the saved tomato exterior. Each has mass and a collision sphere. Elastic bonds link neighboring particles, resisting changes in their separation. The contract describes a 26-neighbor central-force network.

Central-force means a bond's elastic action follows the line connecting its endpoints. The network can resist more directions through many such bonds, but its aggregate material response is determined by that network. Assigning a nominal Young's modulus does not make every continuum property independently adjustable.

31.3 Why the Poisson response changed

Poisson behavior describes lateral deformation when a material is stretched or compressed. The v28 contract reports an effective small-strain value of 0.25, compared with 0.4 in the old reference. The network therefore does not reproduce that reference's full constitutive behavior even before fracture.

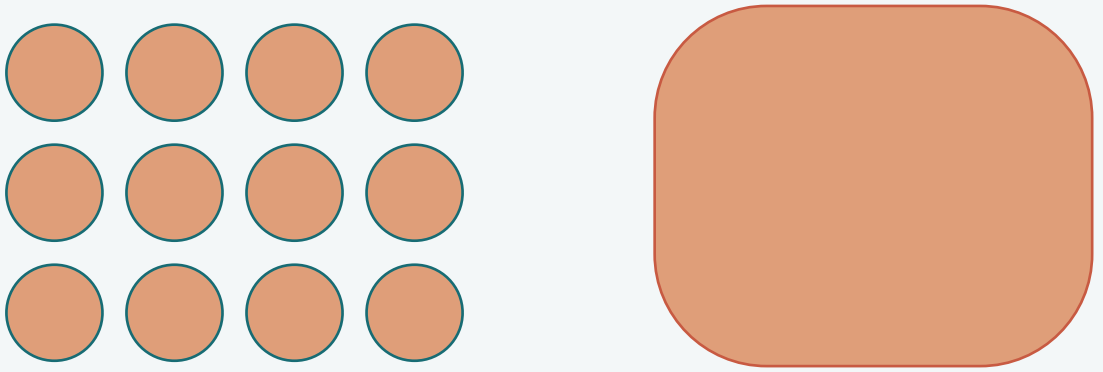
The measured imposed-strain differences, reported as roughly 4-66% across sampled comparisons, make that limitation concrete. This is why 'same nominal stiffness' would be an incomplete description of the approximation.

31.4 Spheres and bonds are not a filled continuum

The collision spheres occupied only about 38% of the previous continuum volume. The bonds transmit force but have no collision volume themselves. A smooth rendered surface can cover those spaces visually without changing the physical gaps.

This model therefore checks particle-level penetration and contact under its own representation. It cannot inherit a guarantee that a blade never crosses any point of the entire imagined tomato volume. The distinction is essential when interpreting the small reported penetration number.

EXPLANATORY SCHEMATIC — not an experimental result



Sphere contact leaves gaps

Smooth display can hide those gaps

31.5 The blade proxy is intentionally different

The contract describes a tapered triangular prism with a 120 mm edge, 30 mm height and 1 mm spine thickness, with an ideal sharp edge. It replaced a failed blunt-box proxy. It is not the original 74-vertex CAD blade.

Tapering matters because a blunt box can require excessive pushing before the intended fracture mechanism activates. But making a proxy more useful for the task does not prove it matches a real kitchen blade. The change and its purpose must remain visible.

31.6 What makes a bond eligible to fail

The approximate fracture law combines geometric intersection with a finite edge/advance process zone and accumulated work. A connection does not disappear solely because a future plane was supplied. The blade has to reach eligible material and spend the prescribed budget.

Eligibility is still a model-specific rule. An edge contacting a bond is not a direct microscopic account of tissue rupture. It is the chosen approximation that links tool movement to irreversible connection loss.

31.7 Why sawing contributes in this model

The recorded correction allowed fracture progress from positive edge-normal movement plus a weighted contribution from edge-tangent slip. The provisional tangential weight is 0.25. Resistance follows total relative movement while maintaining the model's Gc times area budget. [S20]

This was an empirical, uncalibrated law, not a measured discovery about real tomato sawing. It helped the restricted task represent slicing without requiring all progress to come from normal penetration. Its behavior needs validation if used to support a physical claim about cutting strategy.

31.8 The area assigned to a broken bond is approximate

The contract uses a bond-area assignment based on lattice spacing, exact in its stated planar-grid count for axis-aligned cuts. Oblique directions produce different crossings, so directional errors had to be measured.

A network can therefore have a preferred direction even when the displayed tomato is round. The reported directional fracture-area error and failed oblique case are meaningful limitations, not cosmetic details that a smoother exterior repairs.

31.9 Explicit stepping, retry and hard stops

The nominal physics step was 0.2 ms, with finer 0.1 and 0.05 ms options. A complete-state rollback could retry certain numerical or particle-penetration failures with a smaller step, retaining the finer step afterward. Force-stop trials were not bypassed by this mechanism.

This means runtime depends on the encountered state. A hard case may slow down. A benchmark on passing nominal cases does not guarantee the same rate for every angle, size or action sequence.

31.10 Why the restricted pass was still valuable

The benchmark included actions, observations, connectivity, rewards, termination and selective resets, and the eight bodies completed cuts and withdrawal. This was much closer to a learning environment than a fixed-topology force-kernel timing.

It provided a concrete place to begin a restricted control study. The next scientific challenge is to show that useful behavior survives held-out conditions and stronger evaluation, rather than exploiting the model's gaps, network directionality or empirical work law.

32. From the material solver to the robot in the kitchen

32.1 The tool pose is an interface variable

The robot engine knows the blade's native position and orientation. The material solver needs that information in its own coordinate convention. The renderer may use yet another scene hierarchy. A transform mistake can make contact appear at an offset or reverse a direction.

Interface checks tested force, moment, power and clocks. A dry route tested whether the actual robot could traverse the planned path. Neither test could replace the coupled cutting run, but both reduced ambiguity when that run failed. [S16, S21]

32.2 Why actual-pose replay was informative

The blade trajectory recorded from the robot was replayed through the material solver independently. Similar final broken connections and material behavior supported the interpretation that the solver was responding to the actual route rather than a hidden desired-motion shortcut.

Replay is not experimental validation and does not remove all coupling sensitivity. It is a useful consistency check between the integrated and isolated paths. The saved report records its own force and position differences rather than claiming exact equality of every output.

32.3 The one-step delay and action-reaction mapping

Material reaction was applied to the native tool center of mass one 1 ms interval later. Converting the distributed blade loads to that point requires both resultant force and moment. Applying the same reaction twice would artificially increase resistance.

The interface explicitly applied it once. Even so, a delayed partitioned exchange can differ from a finer exchange or fully coupled solve. The reported force differences across runs did not establish strict force convergence.

32.4 The blade-board incident

One withdrawal trial encountered a large native contact event even though preceding geometric measurements showed positive clearance. The native blade and board each had a speculative contact offset, creating a combined interaction range larger than the actual gap.

The corrected candidate reduced the contact envelope while retaining the blade/board geometry and geometric clearance stop. This did not prove contact offsets are unphysical in general; it addressed the documented configuration conflict. The earlier failed run

remained failed.

32.5 Why settling was part of success

After withdrawal, the pieces should not keep moving excessively or violate support conditions. An early run separated and withdrew but did not reach the required quiet state within its initial waiting period.

A later controller allowed a longer bounded settlement window and still required the same continuously quiet condition. Changing the allowed waiting time is different from weakening the velocity threshold. The successful result describes the complete declared episode, not just the exciting moment when two components appeared.

32.6 What the mount and bottom support conceal

The ideal knife mount removes the problem of fingers slipping under cutting load. Fixed bottom-layer support removes much of the problem of a tomato rolling or being squeezed by a holding hand. Both are useful isolations for development.

They also delimit the result. The demonstrated cutting cycle is not evidence of active dual-arm stabilization or a free grasp surviving cutting. Those interfaces must be qualified before presenting the system as the full original bimanual task.

33. Visible tomato versus physical tomato

33.1 Why the particle view was hard to interpret

A collection of spheres and connections makes the numerical structure visible but does not look like an ordinary tomato. That can make it harder for a viewer to understand where the cut is, which surface is exposed, or whether pieces remain connected.

Version 29 therefore added a smooth visual representation. This was useful communication work, provided the surface remained attached to actual material state rather than becoming an independent animation.

33.2 How the display follows the state

The recorded renderer exposes internal faces when their corresponding physical bond breaks. Nearby particle displacements deform the surface. Interpolation groups split at broken local connections so disconnected fragments are not visually blended together. [S14]

This is stronger than replacing one whole tomato mesh with two prebuilt halves at a scheduled time. It ties the visible split to topology produced by the solver. It is still a display transformation, not a new constitutive law.

33.3 Why cosmetic seeds are not seed mechanics

Real seeds, gel, skin and flesh can differ mechanically. A visual seed does not have its own mass, collision response, material interface or fracture rule unless those are implemented explicitly.

The document therefore describes the colors, seeds, calyx and stem as appearance features in this branch. A screenshot cannot establish anatomical realism. That would require separate modeling and evidence.

33.4 Why pieces can remain close after a successful split

Two disconnected pieces need not fly apart. With gravity, support and little stored energy, they can remain touching or nearly adjacent. Their mechanical independence is determined by remaining material connections and response, not only by a dramatic visible gap.

The integrated final scene did not add an artificial spread to make the cut look larger. A separate optional viewer pull-apart impulse was disclosed where present. Such an impulse can help inspect separation, but it must not be presented as the natural outcome of cutting alone.

33.5 Useful visual checks

Compare the smooth view to raw particles. Ask which broken connections expose a face, whether two pieces share interpolation, and whether the visible blade matches its collider. Inspect support and attachments rather than judging only the central action.

These checks do not replace numerical validation, but they help catch mismatches between the story a video tells and the model that generated it.

34. Speed results translated into practical meaning

34.1 Relative speedup can still be unusably slow

An algorithm that improves from 50 seconds to one second per simulated millisecond is fifty times faster. Yet simulating one second still costs about one thousand real seconds. The relative improvement is genuine, but the absolute rate may remain unsuitable for learning.

This pattern appeared in the reference-mechanics GPU work. Large software speedups should be celebrated as implementation progress without being confused with a full-task throughput pass.

34.2 Bulk-only equivalents are a planning number

A benchmark can advance material by 20 ms and call that a 50 Hz transition equivalent. If it omits blade contact, cutting updates, observations, reward, termination, robot dynamics and reset, it has not actually executed the eventual learning step.

The v19 and v22 reports explicitly separated these costs. Their rates were useful for identifying whether enough computational headroom remained, not for promising a complete training duration.

34.3 Interpreting the approximate benchmark

The v28 measured rate was about 136.8 task transitions per wall second across eight environments. Because each task used a 20 ms control interval, that describes aggregate independent task advances, not a single robot operating at 136.8 Hz.

Illustration derived from that rate: 1,000,000 transitions divided by 136.8 is about 7,310 seconds, or 2.03 hours, before adding excluded policy-learning and robot costs or accounting for different cases. This is arithmetic, not a forecast of successful training or convergence.

34.4 Why the native robot result is slower

The integrated robot adds articulation stepping, tool exchange, collision handling, scene operations and potentially rendering. Startup, pickup and approach also consume time. The report's 366-second headless section did not include every part of an entire application launch.

The graphical run took longer still. Comparing either number directly to a material-only kernel or to a batched isolated task would mix different workloads. A meaningful training forecast must measure the exact environment and observation pipeline being trained.

34.5 Resets and failures affect the average

If a hard episode retries at a finer step, stalls, fails early or requires a fresh process, its cost differs from a nominal completed episode. Frequent image rendering or model inference also changes throughput.

A useful future report should include the distribution of episode durations and failure handling, not just one warm best-case rate. It should distinguish wall time, simulated duration, aggregate transitions and learning-update time.

35. How to decide what to trust in a result

35.1 Start with the question the test actually asked

Was it testing geometry, mechanics, implementation equivalence, timestep sensitivity, contact, fracture, reset, perception or task success? A passed result is strongest when the question is narrow and explicit.

The 47-patch energy sweep answered a different question from a live knife cut. A saved-image error measurement answered a different question from an unsupported grasp. The project became more understandable when those questions were separated.

35.2 Positive tests and negative controls

A positive test should succeed under supported conditions. A negative control deliberately creates an invalid or unsupported condition and checks that the system refuses it. Examples include deep penetration, an uncovered interface vertex, a blank camera frame or an unsupported extra cut slot.

A negative control passing means the rejection worked. It does not mean the invalid task was achieved. Large assertion counts often include such controls and bookkeeping checks; they are not counts of independent scientific trials.

35.3 Independence requires more than another filename

An independent test should not have been used to tune the candidate. Renaming a failed placement and trying it again after adjusting the controller makes a useful regression test, but it does not restore its independence.

The pickup sequence reserved fresh test sets after earlier failures guided changes. That preserves an honest interpretation of later success rates. Combining selected successes from multiple controller versions would give a misleading number.

35.4 Internal agreement versus reality

CPU/GPU agreement checks whether two implementations produce compatible results. Refinement checks whether numerical resolution changes the answer. Analytical tests compare against known solutions in restricted fixtures. Experimental validation compares to measured reality.

All are valuable. None automatically supplies the others. An uncalibrated model can match its CPU reference beautifully and still predict the wrong tomato forces.

35.5 Source hashes make claims traceable

A hash identifies exact file contents. Saved configurations and source hashes connect a result to the code and parameters that generated it. This is especially important when the active controller changes repeatedly while experiments continue.

A hash does not certify scientific correctness. It makes the evidence inspectable and helps prevent accidental mixing. Publication manifests likewise verify archived bytes without making live snapshots into an atomic completed experiment.

35.6 A compact reading checklist

For any headline number, identify the model version, geometry, support, blade representation, force definition, timestep, sample count, success condition, omitted costs and failed controls. Then ask whether the claimed conclusion stays inside that envelope.

This approach is more useful than categorizing an entire branch as simply 'good' or 'bad'. A failed full objective can contain valuable verified components, and a successful demo can retain important untested assumptions.

36. A complete imaginary cut, explained moment by moment

36.1 This is a teaching walkthrough

The following sequence is conceptual. It combines the questions a complete cutting environment should answer, rather than claiming that every project branch implemented them identically. Where our actual results are narrower, the earlier chapters and source reports define the boundary.

Imagine a supported tomato, a knife held above it, and a controller choosing a small sawing-and-feed action. The simulator begins from an intact saved state with zero new fracture work.

36.2 Approach and first contact

As the blade approaches, broad-phase search finds nearby material. Narrow-phase queries locate actual contact and choose normals. Before contact, a proper no-contact control should not produce a cutting reaction or damage.

At first contact, material deforms and normal reaction develops. A blunt face may press without being eligible to cut. The reaction sent to the robot must correspond to the contact that actually occurred, not the desired blade position.

36.3 Indentation before rupture

Some tool work is stored elastically. The tomato may move against its supports. Local volume and penetration checks prevent accepting an impossible fold or deep crossing.

If force exceeds the declared stop before valid fracture, the episode fails or stops. The simulator must not break bonds merely to avoid that outcome. Some project candidates stopped precisely at this stage, which is why they were not called completed cuts.

36.4 Damage begins and a partial bridge remains

The model's eligibility and work rule permits a small region to damage or fail. New motion freedom allows the lips to open there. Material ahead of the front remains connected and continues transmitting force.

The history records what has happened. If the blade reverses slightly, the damage should not disappear. If the solver retries a rejected numerical trial, the damage from that rejected guess should not survive either. Physical irreversibility and transactional rollback coexist by referring to accepted history.

36.5 The final bridge fails

Eventually the remaining material path between the intended pieces disappears. A connectivity check finds more than one component. The fracture ledger accounts for the newly failed area under the chosen law.

This is a necessary separation event, but further checks remain: the pieces must not still share hidden basis variables or grid transfers, and the knife must leave without interpenetration or an invalid contact response.

36.6 Withdrawal and recontact

The knife retreats. Contact changes and elastic energy may be released. The pieces may touch each other again. That contact should repel overlap without restoring the old cohesive connections.

A piece can remain close to its neighbor because of support and gravity. Independence is tested mechanically, not by requiring a large visual gap. The pulse-control idea is useful here when designed with the appropriate baseline and external loads.

36.7 Termination and reset

A complete episode checks clearance, support, quietness and other declared success conditions. It records failures even when an earlier stage looked successful. Then reset restores the intended intact starting state, including connections and work counters.

For a batch, resetting one episode must leave the others unchanged. For a coupled robot, restoring material alone is not an exact reset of every native contact state. Those distinctions explain why the project accumulated many reset tests rather than a single universal reset checkbox.

37. Failure walkthrough: the blade passed between the checks

37.1 The problem

The early full-cut continuation had accepted shallow states, but a finite-triangle counterexample revealed a hole in the contact test. The blade could intersect the interior of a material face while the selected vertices remained outside. The solver had not necessarily failed to solve its equations; the equations were missing a geometric constraint. [S04]

37.2 An everyday analogy

Imagine inspecting a fence only at its posts. Finding every post intact does not prove that nothing passed through the space between them. Similarly, testing mesh vertices tells us about those vertices, not every point of the triangular material between them.

This analogy explains why 'more accurate vertex distances' alone would not repair the coverage problem. The test needed to include the finite surface, not merely improve arithmetic at the same limited sample locations.

37.3 What changed

The new offline backend introduced finite-triangle contact constraints and weighted force handling. The saved counterexample became a regression target: it should now be detected and rejected or resolved under the corrected formulation.

The deeper nonlinear continuation still had its own difficulties. Fixing geometry removed one false sense of safety; it did not guarantee that the coupled damaged material could be advanced cheaply or robustly to release.

37.4 What to take from this case

When a report says 'penetration passed', ask whether it refers to vertices, surface samples, finite triangles, particle spheres or the full material volume. A stricter geometric test can uncover a defect in a previously accepted-looking trajectory without changing the material itself.

38. Failure walkthrough: the material moved but the strain did not

38.1 The problem

The v02 candidate reconstructed a geometric state that became locally invalid even though its stored strain indicator remained near the intact range. An independent transfer analysis found non-rigid modes that moved particles without a corresponding constitutive response. [v02]

This is more serious than a visual glitch. The material's restoring force depends on its strain representation. If that representation misses deformation, the solver can allow motion that should cost elastic energy.

38.2 How the diagnosis isolated the cause

The investigation used an independent small-mode fixture rather than relying only on the failed cutting trajectory. That removed many confounding effects: instead of a complicated blade interaction, it asked whether the transfer could carry a deformation invisibly.

A separate volume certificate caught the invalid geometry. Both pieces of evidence mattered. One showed the state was unacceptable; the other explained a mechanism that could produce it. Simply adding a stricter stop would leave the inability to make useful progress unresolved.

38.3 What the repairs taught us

The APIC-style candidate removed the reproduced mode but still failed the required timestep consistency. The following consistent-field candidate avoided recurring filtering and tied deformation and position to one field, passing its intact checks at the selected small step.

The lesson was not that every earlier component had been worthless. It was that a transfer scheme, strain update and force mapping must be mutually compatible. A successful geometry or force-balance test cannot substitute for that compatibility.

39. Failure walkthrough: the crack existed but could not open

39.1 The problem

Version 25 reached and paid a shallow interface, yet the partial crack had no independent opening mode. The interface geometry and energy accounting existed, but the displacement representation still tied the two lips together. A complete-plane control could open. [v25-v26]

39.2 Why this was easy to misunderstand

The phrase 'fracture event occurred' sounds like separation has been achieved. In reality, an event can update an area or history record without providing the mechanical freedom needed to realize an opening.

Imagine drawing a slit on a sheet but leaving a transparent backing holding both sides. The slit is visible, but the backing still constrains movement. In the numerical model, the hidden backing was shared continuity in the movement representation, not a physical extra layer.

39.3 The targeted correction

Version 26 changed local continuity so the crack lips could move independently behind the front. Tests checked partial depths, full-plane behavior, mass, affine fields and restart. A tiny CAD-driven work-paid crack then advanced with scoped timestep comparisons.

This was a real improvement, but the episode was still a tiny corner-crack test. It did not establish a whole tomato slice. The representation grew during the advance, and repeated geometry/operator work remained expensive.

39.4 Why this case matters for the paper

If a contribution concerns cutting, topology labels alone are not enough. The paper must show that the mechanical system acquires the right freedoms while retaining the intact bridge. A convincing video should be accompanied by a test that distinguishes a genuinely opening crack from an updated display or ledger.

40. Failure walkthrough: stable did not mean accurate

40.1 The problem

Broad aggregation improved the timestep-related stability diagnostic around tiny corner-cut regions, but the resulting dynamic response differed substantially from the original reference. Some initial projection checks looked good, and both methods could converge with their own smaller steps. [v18]

40.2 Why those passes did not rescue it

Projection asks whether the current state can be represented without a large immediate jump. Dynamics asks how that representation responds over time. A change can preserve today's shape while changing tomorrow's acceleration and vibration.

Likewise, each method can be internally converged yet disagree with the other because it solves a different discrete model. Reducing the timestep then makes both answers more precise without bringing them together.

40.3 What changed next

Selective aggregation limited the restriction to very weakly supported regions. The accepted comparison used an explicitly approved mass-weighted velocity measure. The report continued to show larger pointwise errors rather than implying the original maximum-point criterion had passed.

This is a legitimate scoped decision only when disclosed. The reader needs to know what accuracy means in the accepted result. A later contact study must still consider whether a small affected region has outsized importance near the blade.

40.4 The practical lesson

Whenever a speed technique changes the space of allowed motion, it deserves physical-response comparisons, not only faster runtime and finite values. Pure software reuse and numerical approximation should be distinguished, even when both are called optimizations in conversation.

41. Failure walkthrough: better vision did not guarantee pickup

41.1 The problem

The expanded camera model located grasp targets within a few millimetres on saved validation images, but the heavier knife on its opposite face still encountered slips and force spikes in live pickup. Teacher-target feasibility and camera-driven task success were different records. [S17, S19]

41.2 Why a small error can matter

A small shift on a flat wide surface may be harmless. The same shift near a sloping handle edge can move a fingertip's force direction, shorten its lever arm or cause another finger to push first. The camera error number does not describe the entire contact tolerance of the grasp.

The heavier blade also changes the turning moment the grip must resist. Accurate perception of the handle does not automatically choose a force-balanced configuration. The task therefore required a useful target and a controller capable of establishing and retaining contact there.

41.3 Why tightening harder failed

An early increase in holding effort worsened a slip. The saved interaction suggested twisting rather than simply insufficient normal pressure. Contact-guided closure and supported balancing improved specific cases by changing the sequence and distribution of loads.

The later difficult pose still resisted those improvements. Smaller steps and another constraint solver did not by themselves remove the spikes in the latest inspected chat. The next work returned to geometry and which opposing fingers could share the load.

41.4 What a successful evaluation must keep separate

Saved-image accuracy measures perception. Supplied-target tests measure controller feasibility. Camera-driven tests measure their combination. Holding for three seconds measures a particular unsupported condition, not retention under all cutting loads.

The first 7/8 result belongs to a frozen short-knife setup. It must not be combined with later selected development successes to manufacture a broader success rate. The reserved final cases should remain a test of one recorded candidate and its settings.

42. A worked guide to the numbers in our reports

42.1 Reading insertion depth

A depth value says where the blade or insertion coordinate reached in a particular trajectory. It does not count remaining material connections. The old 9.638916015625 mm checkpoint still had one connected component and 42 active patches.

The meaningful companion values are damage history, connectivity, contact validity, work balance and the branch's predecessor state. Depth is progress along a coordinate, not a universal measure of task completion.

42.2 Reading a component count

Two components means the tested material connectivity graph has split into two groups. It does not alone prove absence of hidden grid coupling, correct recontact or physically plausible fracture resistance. Those need their own checks.

For the approximate robot demonstration, the final two components were accompanied by broken-bond history, actual-pose replay, withdrawal and settling evidence. That is stronger than a component count alone, but remains inside the model's approximate material and support assumptions.

42.3 Reading a percentage difference

A percentage needs a denominator and a meaningful signal scale. Near zero, a tiny absolute difference can become an enormous percentage. Reports sometimes use a declared absolute floor to prevent misleading comparisons.

Do not remove an inconvenient large percentage without examining that definition. Conversely, do not claim a physically large error merely because a ratio is large when both values are tiny. The raw values and the frozen comparison rule belong together.

42.4 Reading millimetres, micrometres and nanometres

One millimetre is one thousand micrometres and one million nanometres. A very small numerical position discrepancy may reflect two implementations of the same synthetic equations. It does not establish real-world accuracy at that scale.

For example, exact CAD-query agreement with an independent reference is evidence about a geometry calculation. It says nothing by itself about whether the tomato stiffness, blade profile or camera calibration matches reality.

42.5 Reading 'all tests passed'

A test suite may include dozens of assertions on units, array shape, rollback and negative cases. That is useful engineering coverage. It is not the same as dozens of independent physical tomatoes or independent training seeds.

When writing the paper, state the number of trials for the claimed experimental outcome separately from the number of software checks. This prevents a large test count from being mistaken for statistical evidence.

43. From approximate simulation to a defensible cutting study

43.1 Freeze a question small enough to answer

A focused question could ask whether force feedback improves a restricted cutting task under specified variation. It would need a baseline controller, a clear action space, valid observations, a declared model and held-out conditions.

That question is narrower than claiming that the entire material model is realistic or that a robot can cut every tomato without calibration. A useful paper can make a bounded claim if its evidence and limitations are explicit.

43.2 Compare the right controllers

The fixed-motion baseline shows what happens without adaptation. A conventional feedback controller tests whether learning adds value beyond ordinary force regulation. Observation ablations test whether a force channel contributes rather than merely increasing model complexity.

Those comparisons should use the same environment, evaluation conditions and success definition. A learned controller should not be given easier geometry, extra hidden state or a looser stop than its baseline without disclosure and justification.

43.3 Evaluate behavior, not reward alone

A reward is a training signal that may combine many objectives. A high reward can hide an undesirable strategy if the reward or environment has a loophole. Report task completion, load behavior, time, deformation, invalid contacts and failure modes separately.

In this project, model directionality, particle gaps and timestep-sensitive force peaks are candidate sources of misleading behavior. Tests outside the tuned nominal path are necessary to see whether a policy relies on them.

43.4 Separate material fitting from control evaluation

If real measurements are used to fit stiffness or fracture resistance, reserve different specimens or trials to evaluate the fitted model. If controller settings are tuned on a knife pose, reserve other poses for evaluation.

The same principle applies at multiple levels. Reusing one dataset for every decision can produce an impressive fit without showing robustness. The report should say which evidence selected the model and which evidence tested it afterward.

43.5 The next physical measurements should disambiguate causes

A cutting-force curve alone combines bulk deformation, fracture, contact, friction, blade geometry and motion. Several combinations of parameters may produce a similar curve. Additional indentation or controlled loading can help separate those effects.

The project saved a data-needed list including specimen geometry, blade motion, sensor axes, calibration, tissue condition and repeated trials. That list explains why extracting a few points from a published graph was kept provisional. [S07]

43.6 What remains a research choice

The team can prioritize the approximate controller study while retaining strict numerical references, or make simulator-method development the main paper. It should not silently switch the claim midway while using old component passes as universal validation.

The expanded document supports that choice by showing what each branch actually established. It does not choose a publication venue, promise an acceptance outcome or declare scientific novelty without the necessary comparison work.

44. The tomato model in plain-language questions and answers

44.1 Is our tomato a real tomato inside the computer?

It is a mathematical approximation. Different versions approximate different features. The strict reference work emphasized consistent deformation, contact and fracture accounting; the fast branch deliberately simplified those details to make restricted task experiments practical. Neither currently establishes all the behavior of biological tomato tissue.

44.2 Have we actually cut it?

Yes, restricted work-paid cuts completed in the approximate model, including a coupled fixed-mount robot demonstration with withdrawal and settling. No, that does not mean the older strict model passed all its unfinished gates. Both statements can be true because they describe different models and evidence.

44.3 Can it be cut anywhere?

No general guarantee is established. Some earlier models prepared a seam or restricted local band. The approximate model does not require a supplied future fracture plane, but an oblique case failed and lattice directionality remains. Freedom to request an action is different from demonstrated robust success for that action.

44.4 Does the smooth skin change the forces?

Not in the documented visual upgrade. The skin, seeds and flesh colors improve interpretation of the display. Collision and material behavior remain governed by the underlying approximate representation. A future anatomical model would require additional physical laws and evidence.

44.5 Has the robot learned to cut?

The record does not establish a trained adaptive cutting policy. It includes resource PPO exercises, scripted cutting, a restricted learning-ready task interface and learned visual grasp targeting. These are foundations for learning, not the completed original learning claim.

44.6 Why did we keep the older models?

They preserve diagnoses, reference equations, acceptance tests and rejected assumptions. They help explain which shortcuts the fast model makes and provide comparisons for later work. Keeping them does not require continuing every branch indefinitely.

44.7 What should I remember most?

Always name the model and the test. 'The approximate model completed this supported cut' is informative. 'The tomato is finished' hides too many unresolved questions. The project has delivered meaningful components and demonstrations; its next strength will come from connecting them under a focused, honestly evaluated research question.

Appendix A. Glossary

APIC

Affine Particle-In-Cell: a particle/grid transfer approach carrying local velocity variation. The project tested a particular adaptation; its failure does not invalidate the entire method.

Basis / coefficient

A set of building-block motion shapes and their weights. The basis decides what deformations the model can express.

Calibration

Choosing model parameters using measured data. Separate measurements are needed to validate the fitted model.

CAD / collision geometry

The detailed authored shape and the shape actually used to compute contact. They may differ from the rendered display.

Cohesion

Resistance holding opposing material faces together before complete failure.

Connectivity / component

Which material points remain joined. A connected component is a group reachable through remaining material connections.

Consistent mass

An inertia description that retains coupling implied by the displacement representation. A lumped mass is a deliberately simpler local approximation.

Convergence

Checking whether results approach agreement as numerical resolution is refined. Solver convergence, timestep convergence and mesh convergence are different checks.

Coupon

A small specimen designed to isolate a particular physical or numerical behavior.

Damage history

A remembered state describing how much cohesion has been irreversibly lost. With no healing, unloading does not reset damage.

Deformation gradient / J

A local description of how material changes shape. Its determinant J measures local volume ratio; a negative value indicates inversion.

Domain randomization

Varying simulation conditions during training to discourage dependence on one exact configuration. It does not by itself prove realism.

FEM

Finite Element Method: approximate a continuous material using elements and a displacement field, then derive forces and motion.

Fracture energy

Work needed per new cut area in a declared material law. A correct energy counter is not experimental proof of that law.

Gate

A specific acceptance condition that must pass before the next stage. Passing another unrelated check does not override it.

GPU synchronization

Waiting for GPU work to finish before the CPU uses the result. Frequent transfers can make a nominal GPU implementation slow.

Ground truth / teacher target

Exact or deliberately supplied information used for labels or development. It must not silently replace camera inference in an image-only test.

Impedance / compliance

How motion responds to force, often described with spring-like stiffness and damping. Compliance means yielding under load.

Impulse / wrench

Impulse is force accumulated over time. A wrench is force and torque together, expressed at a specified point and frame.

MPM

Material Point Method: material information is carried by points and transferred through a grid. Preventing unintended sharing across cut pieces is a central issue.

Null mode / rank defect

A motion that the discrete model fails to resist or measure as intended. Rank checks help reveal missing mechanical constraints.

Observation mask

A rule limiting which simulator quantities a controller is allowed to see.

PPO / reinforcement learning

PPO is one algorithm for improving a policy using rewards from interaction. Running PPO software does not establish a particular learned skill.

Quadrature

A numerical weighted-sampling rule for integrating force, mass or energy over a region. New thin regions can require new sampling.

Quasistatic

Finding a sequence of approximately force-balanced states while neglecting the full transient dynamics.

Residual policy

A learned correction added to a baseline controller, rather than learning every motion from scratch.

Rollback / transaction

A trial step is accepted as a whole or restored to its previous state, including history and counters.

Simulated time / wall time

Time inside the model versus actual elapsed time on the laptop. Their ratio can be far from one.

Stiffness / damping

Resistance to deformation versus resistance associated with motion. Adding damping can hide instability while changing behavior.

Transition throughput

How many complete environment action steps execute per actual second. Material-only equivalents are not full robot transitions.

Validation / independent test

Validation guides model selection; an untouched test evaluates the frozen choice. Once used for tuning, a test is no longer independent.

Work / energy ledger

A record of input work, stored energy, motion energy and dissipation used to expose missing or double-counted effects.

Appendix B. Version-by-version guide

The v01-v33 identifiers below refer to the isolated November experiment folders. Earlier DiSECt work also used version numbers, so a bare version number without its folder can be ambiguous. This guide preserves that distinction.

v01 - Persistent history

The candidate remembered geometric cuts after withdrawal and passed restricted fragment-independence controls. It had not yet validated physical fracture, contact or useful complete-loop speed.

Evidence: tomato_persistent_mpm_20261110_v01/REPORT.txt

v02 - Physical qualification failure

Reconstructed material could invert while stored strain stayed near its initial value. Finer surface sampling and a whole-volume guard detected the problem; the guard did not cure it.

Evidence: tomato_persistent_mpm_gate3_20261110_v02/REPORT.txt

v03 - APIC transfer repair

The specific invisible-motion control was repaired. Repeated transfer damping and timestep inconsistency still prevented physical qualification.

Evidence: tomato_apic_repair_20261110_v03/REPORT.txt

v04 - Consistent intact mechanics

One displacement field described geometry and strain, with full consistent inertia. Restricted intact tests passed at a very small timestep. Cut geometry and blade contact still needed requalification.

Evidence: tomato_consistent_mechanics_20261110_v04/REPORT.txt

v05 - Restricted cut-aware basis

A partial cut could open while preserving its bridge, and a full division allowed independent pieces. Moving a cut off the sampling grid exposed unsupported numerical modes.

Evidence: tomato_cut_basis_20261110_v05/REPORT.txt

v06 - Cut-aware integration

Integration over cut-created subcells repaired the tested off-grid defect without adding mass or changing material stiffness. It remained a restricted prescribed-topology block test.

Evidence: tomato_cut_integration_20261110_v06/REPORT.txt

v07 - Multiple cuts

Supplied parallel and cross-cut fixtures produced additional independent fragments and retained history. No blade work caused these topology events, and manufactured gaps were used in independence checks.

Evidence: tomato_multicut_20261110_v07/REPORT.txt

v08 - First GPU comparison

A straightforward GPU implementation improved reference speed but left force assembly expensive. The later v09 report preserves the comparison; this is not a complete cutting-task rate.

Evidence: see the following version report and experiment source directory.

v09 - Parallel force evaluation

Parallelizing the same numerical sum produced large relative speedups while retaining reference agreement in tested cases. Absolute runtime for a tiny block remained much too slow.

Evidence: tomato_gpu_comparison_20261111_v09/REPORT.txt

v10 - Current blade sweep creates topology

Successive finite cutting-edge poses generated restricted geometric cuts without a future supplied route. Mechanical evolution ran on GPU, but topology rebuilding still used the CPU.

Evidence: tomato_blade_sweep_20261111_v10/REPORT.txt

v11 - Fine crack front and deformed material

Finer front representation exposed fast local modes and a failed velocity-refinement check. Exact planar mapping also rejected a freely evolving non-affine specimen.

Evidence: tomato_deformed_front_20261111_v11/REPORT.txt

v12 - Finite-face deformed-front correction

The finite CAD cutting face permitted a more conservative geometric mapping. Some finer mechanics checks passed, but combined blade-update refinement still failed.

Evidence: tomato_deformed_face_20261111_v12/REPORT.txt

v13 - Separate timing from spatial changes

Controlled replays showed both event timing and changed motion space contributed to disagreement. A diagnostic mesh using future saved data was not advertised as an online solution.

Evidence: tomato_activation_audit_20261111_v13/REPORT.txt

v14 - Local band and event timing

A preselected local interaction band and precisely localized activation passed a restricted comparison. The blade could not freely leave that band or change plane.

Evidence: tomato_event_front_20261111_v14/REPORT.txt

v15 - Performance no-go

Profiling rejected that high-accuracy consistent-mass formulation as the laptop online-learning backend. This was a cost decision, not a universal impossibility result.

Evidence: tomato_performance_review_20261111_v15/REPORT.txt

v16 - Local-state approximation

Local inertia and trilinear fields reduced coupling complexity, but angled cuts produced extra zero-strain modes. Even intact material-only timing missed the entry target.

Evidence: tomato_local_state_20261111_v16/REPORT.txt

v17 - Linear tetrahedral repair

Linear elements repaired the tested angled-cut rank defect. Very small cut regions constrained the timestep, and the component remained too slow for the intended full task.

Evidence: tomato_linear_local_20261112_v17/REPORT.txt

v18 - Broad aggregation

Constraining weak cut-region degrees of freedom improved a stability diagnostic but changed dynamic response too much. Good initial projection did not guarantee accurate later motion.

Evidence: tomato_aggregation_20261112_v18/REPORT.txt

v19 - Selective aggregation

Only the weakest-supported degrees of freedom were constrained. Component agreement passed under an explicitly approved mass-weighted velocity criterion; larger local errors stayed visible.

Evidence: tomato_selective_aggregation_20261112_v19/REPORT.txt

v20 - Incremental geometry and caching

Unchanged geometry and repeated operators were reused, and GPU assembly improved. The changing-geometry loop became faster without becoming a complete learning environment.

Evidence: tomato_incremental_gpu_20261112_v20/REPORT.txt

v21 - Compiled events and persistent GPU buffers

Compiled clipping and reused GPU graphs reduced event overhead. A complete geometric diagnostic still remained below the preliminary throughput target.

Evidence: tomato_compiled_events_20261112_v21/REPORT.txt

v22 - Overlapped changing-geometry episodes

Independent preparation and stepping overlapped. Repeated synthetic geometric episodes exceeded 100 aggregate transitions/s, before physical contact, fracture, robot and policy costs.

Evidence: tomato_cut_update_redesign_20261112_v22/REPORT.txt

v23 - Original-CAD contact adapter

The material and blade were coupled with rollback and conservative volume checks. Contact sampling had not converged, and dense contact was expensive.

Evidence: tomato_contact_integration_20261112_v23/REPORT.txt

v24 - Contact precision and integration

Wrong nearest-face candidates on the thin blade were corrected with original double-precision triangles. Refined contact tests passed for an intact small cube, not a full tomato cut.

Evidence: tomato_adaptive_contact_20261112_v24/REPORT.txt

v25 - Partial crack cannot open

Tomato geometry and contact were integrated, but a work-paid shallow interface still had zero opening rank. A full-plane control could open, exposing an overly constrained partial representation.

Evidence: tomato_cut_surface_20261113_v25/REPORT.txt

v26 - Local continuity repaired

Partial crack lips acquired independent motion. A shallow CAD-driven paid crack advanced with scoped refinement agreement, but only over a tiny interval and at high cost.

Evidence: tomato_local_continuity_20261113_v26/REPORT.txt

v27 - Exact-search optimizations and scope decision

Hierarchical contact/clearance searches and reduced redundant work improved components. The full fast-tomato objective remained unmet, leading to an explicit approximation decision.

Evidence: tomato_bounded_basis_20261113_v27/REPORT.txt

v28 - Approximate training task

A particle-bond model completed restricted work-paid cuts and batched task loops at about 136.8 transitions/s. Its material error, voids, proxy blade and force sensitivity were disclosed.

Evidence: tomato_approx_rl_20261113_v28/RESULT.txt

v29 - Smooth visual surface

The displayed exterior and exposed interior followed material connections. This added visual clarity, not physical tissue layers or continuum collision volume.

Evidence: tomato_surface_view_20261113_v29/README.txt

v30 - Scripted motion comparisons

Faster feeds shortened selected simulated cuts but increased loads in some cases. Smooth feed at reversal could fail; synchronization helped one tested variant.

Evidence: tomato_motion_comparison_20261113_v30/REPORT.txt

v31 - Native robot integration

Measured robot blade motion drove the approximate tomato and received reaction feedback. A complete cutting cycle passed with an ideal fixed knife mount and supported tomato.

Evidence: tomato_robot_integration_20261113_v31/RESULT.txt

v32 - First visual contact-only pickup

Learned camera targeting plus engineered contact control passed 7/8 fresh short-knife tests. Other knife variants and the opposite resting face initially failed.

Evidence: vision_knife_pickup_20261113_v32/README.txt

v33 - Broader visual pickup, ongoing

Three shapes and two faces were explored with more image data. Teacher-target feasibility and saved-image accuracy improved, but difficult live grips remained unresolved.

Evidence: vision_knife_pickup_generalization_20261114_v33/PLAN.txt

Appendix C. Evidence and reading map

These are primary project records, not a newly verified literature review. Links below open local source files on the project laptop. Many are also preserved in the GitHub archive. Earlier plans can contain expectations later overturned by tests; a plan is not evidence of completion.

Published snapshot: TianjinUniversity research archive, verified November 14 commit

S01 - Early research question and decision log

research_question.md
/home/omen/.gemini/antigravity/scratch/TianjinUniversity/research_state/research_question.md

decisions.md
/home/omen/.gemini/antigravity/scratch/TianjinUniversity/research_state/decisions.md

S02 - Phase audit, laptop benchmark and native integration

PHASE_STATUS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/PHASE_STATUS.md

PHASE2_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/PHASE2_RESULTS.md

NATIVE_INTEGRATION_AND_PHASE3.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/NATIVE_INTEGRATION_AND_PHASE3.md

S03 - Early grasp feasibility and load-support diagnostics

CENTRAL_GRASP_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/CENTRAL_GRASP_RESULTS.md

IN_PLACE_TILT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/IN_PLACE_TILT_RESULTS.md

PALM_SEATING_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/PALM_SEATING_RESULTS.md

CONTACT_SEATING_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/CONTACT_SEATING_RESULTS.md

CONTACT_TRANSFER_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/CONTACT_TRANSFER_RESULTS.md

LOAD_SUPPORT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/LOAD_SUPPORT_RESULTS.md

S04 - Offline material, fracture and contact history

TOMATO_4MM_MESH_COMPARISON_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_4MM_MESH_COMPARISON_RESULTS.md

TOMATO_4MM_REMAINDER_REFINEMENT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_4MM_REMAINDER_REFINEMENT_RESULTS.md

TOMATO_ALL_PATCH_ENERGY_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_ALL_PATCH_ENERGY_RESULTS.md

TOMATO_BLADE_DAMAGE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_BLADE_DAMAGE_RESULTS.md

TOMATO_CLOSED_SURFACE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CLOSED_SURFACE_RESULTS.md

TOMATO_CONTACT_CONTINUATION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CONTACT_CONTINUATION_RESULTS.md

TOMATO_CONTACT_COUPON_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CONTACT_COUPON_RESULTS.md

TOMATO_COUPLED_CONTACT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_COUPLED_CONTACT_RESULTS.md

TOMATO_CUBIC_PATH_DIAGNOSTIC_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CUBIC_PATH_DIAGNOSTIC_RESULTS.md

TOMATO_CURVATURE_SEARCH_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CURVATURE_SEARCH_RESULTS.md

TOMATO_FIRST_FAILURE_STEP_COMPARISON_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_FIRST_FAILURE_STEP_COMPARISON_RESULTS.md

TOMATO_FULL_CUT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_FULL_CUT_RESULTS.md

TOMATO_GLOBAL_SEARCH_DIAGNOSTIC_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_GLOBAL_SEARCH_DIAGNOSTIC_RESULTS.md

TOMATO_INTERFACE_COMPRESSION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_INTERFACE_COMPRESSION_RESULTS.md

TOMATO_LARGE_DEFORMATION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_LARGE_DEFORMATION_RESULTS.md

TOMATO_LAYOUT_AND_BOND_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_LAYOUT_AND_BOND_RESULTS.md

TOMATO_PATCH_ENERGY_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_PATCH_ENERGY_RESULTS.md

TOMATO_SEARCH_RELEASE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SEARCH_RELEASE_RESULTS.md

TOMATO_SLICE_MESH_REFINEMENT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SLICE_MESH_REFINEMENT_RESULTS.md

TOMATO_SPECTRAL_RECLOSURE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SPECTRAL_RECLOSURE_RESULTS.md

TOMATO_SPECTRAL_WITHDRAWAL_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SPECTRAL_WITHDRAWAL_RESULTS.md

TOMATO_SURFACE_MECHANICS_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SURFACE_MECHANICS_RESULTS.md

TOMATO_SWEPT_PATH_DIAGNOSTIC_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SWEPT_PATH_DIAGNOSTIC_RESULTS.md

TOMATO_SWEPT_REFINEMENT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_SWEPT_REFINEMENT_RESULTS.md

TOMATO_TRIANGLE_CONTACT_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_TRIANGLE_CONTACT_RESULTS.md

TOMATO_TRIANGLE_WITHDRAWAL_RECLOSURE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_TRIANGLE_WITHDRAWAL_RECLOSURE_RESULTS.md

TOMATO_UNLOADING_RECLOSURE_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_UNLOADING_RECLOSURE_RESULTS.md

TOMATO_VOLUME_PROTECTION_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_VOLUME_PROTECTION_RESULTS.md

TOMATO_VOLUME_SENSITIVITY_RESULTS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_VOLUME_SENSITIVITY_RESULTS.md

TOMATO_CAPABILITY_AUDIT.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/TOMATO_CAPABILITY_AUDIT.md

S05 - Bounded preparation and preserved checkpoint

SOL_PREUPGRADE_15H_PLAN_20261028.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/SOL_PREUPGRADE_15H_PLAN_20261028.md

session.json
/home/omen/teamwork_projects/tomato_cutting_right_arm/runs/offline/tomato_preupgrade_20261027T171031Z/session.json

PROGRESS.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/runs/offline/tomato_preupgrade_20261027T171031Z/PROGRESS.md

S06 - Native model integration

physics-materials-integration-20261101.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/physics-materials-integration-20261101.txt

S07 - Research milestone review

RESEARCH_ROADMAP_REVIEW_20261102.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/RESEARCH_ROADMAP_REVIEW_20261102.md

tomato-experiment-data-needed-20261101.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/tomato-experiment-data-needed-20261101.txt

S08 - Research slicing extension

research-slicing-extension-20261106.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/research-slicing-extension-20261106.txt

research-slicing-continuation-20261106.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/research-slicing-continuation-20261106.txt

S09 - Solver feasibility profiling

solver-comparison-20261108.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/solver-comparison-20261108.txt

S10 - DiSECt-derived completion and later status

disect-completion-20261108.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/disect-completion-20261108.txt

disect-section4-current.txt
/home/omen/Documents/Codex/2026-10-31/tomato-project-handoff/outputs/disect-section4-current.txt

DISECT_SECTION5_EXECUTION_PLAN_20261110.md
/home/omen/teamwork_projects/tomato_cutting_right_arm/docs/rl/DISECT_SECTION5_EXECUTION_PLAN_20261110.md

S11 - Public CRESSim feasibility

REPORT.txt
/home/omen/Documents/Codex/experiments/cressim_20261110/REPORT.txt

S12 - Strict numerical sequence and scope decision

REPORT.txt
/home/omen/Documents/Codex/experiments/tomato_bounded_basis_20261113_v27/REPORT.txt

NEXT_SCOPE_DECISION.txt
/home/omen/Documents/Codex/experiments/tomato_bounded_basis_20261113_v27/NEXT_SCOPE_DECISION.txt

S13 - Approximate training model

RESULT.txt
/home/omen/Documents/Codex/experiments/tomato_approx_rl_20261113_v28/RESULT.txt

S14 - Smooth appearance

README.txt
/home/omen/Documents/Codex/experiments/tomato_surface_view_20261113_v29/README.txt

S15 - Motion comparison

REPORT.txt
/home/omen/Documents/Codex/experiments/tomato_motion_comparison_20261113_v30/REPORT.txt

S16 - Integrated fixed-mount robot

RESULT.txt
/home/omen/Documents/Codex/experiments/tomato_robot_integration_20261113_v31/RESULT.txt

S17 - Completed visual pickup evaluation

HYBRID_V07_RESULT.txt
/home/omen/Documents/Codex/experiments/vision_knife_pickup_20261113_v32/reports/HYBRID_V07_RESULT.txt

S18 - Repository archival update

PUBLICATION_UPDATE_20261114.json
/home/omen/teamwork_projects/tianjin-tomato-sync/research/simulation/tomato_cutting_right_arm/PUBLICATION_UPDATE_20261114.json

REPOSITORY_SNAPSHOT.md
/home/omen/teamwork_projects/tianjin-tomato-sync/research/simulation/tomato_cutting_right_arm/REPOSITORY_SNAPSHOT.md

S19 - Latest ongoing pickup branch

PLAN.txt
/home/omen/Documents/Codex/experiments/vision_knife_pickup_generalization_20261114_v33/PLAN.txt

S20 - Approximate model: exact declared contract

contract.json
/home/omen/Documents/Codex/experiments/tomato_approx_rl_20261113_v28/contract.json

S21 - Native coupling and failed integration trials

INTEGRATION_NOTES.txt
/home/omen/Documents/Codex/experiments/tomato_robot_integration_20261113_v31/INTEGRATION_NOTES.txt

Conversation evidence and scope

The continuation chat titled “Tomato cutting — repository upload and…” (local chat ID 01a191a6-498e-7a40-be2f-fe0df4a0420c) was inspected for the latest pickup work. Its newest turn was still in progress. The latest inspected message described testing load sharing across opposing fingers after alternative-solver and finer-step trials still produced force spikes. This document does not assume those subsequent tests passed.

The current chat supplies the user choices about publication scope, paper-direction discussion, and verified push outcome. The entire lifetime of every chat was not re-read message by message. The account combines those conversations with the accessible saved result documents. Missing dates or undocumented early actions are not reconstructed as facts.

How to use this appendix

Start with the chapter matching your question. Open its source label here, then the result report. If reproducing a number for the paper, follow that report to the raw run, configuration and source hash; check that the quoted case, timestep, controller and model all match. Do not launch old scripts into an existing evidence folder, because some use fixed output names.